

Microsoft COCO: Common Objects in Context

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Abstract—We present a new dataset with the goal of advancing the state-of-the-art in object recognition by placing the question of object recognition in the context of the broader question of scene understanding. This is achieved by gathering images of complex everyday scenes containing common objects in their natural context. Objects are labeled using per-instance segmentations to aid in precise object localization. Our dataset contains photos of 91 object types that would be easily recognizable by a 4 year old. With a total of 2.5 million labeled instances in 328k images, the creation of our dataset drew upon extensive crowd worker involvement via novel user interfaces for category detection, instance spotting and instance segmentation. We present a detailed statistical analysis of the dataset in comparison to PASCAL, ImageNet, and SUN. Finally, we provide baseline performance analysis for bounding box and segmentation detection results using a Deformable Parts Model.

1 INTRODUCTION

One of the primary goals of computer vision is the understanding of visual scenes. Scene understanding involves numerous tasks including recognizing what objects are present, localizing the objects in 2D and 3D, determining the objects' and scene's attributes, characterizing relationships between objects and providing a semantic description of the scene. The current object classification and detection datasets [1], [2], [3], [4] help us explore the first challenges related to scene understanding. For instance the ImageNet dataset [1], which contains an unprecedented number of images, has recently enabled breakthroughs in both object classification and detection research [5], [6], [7]. The community has also created datasets containing object attributes [8], scene attributes [9], keypoints [10], and 3D scene information [11]. This leads us to the obvious question: what datasets will best continue our advance towards our ultimate goal of scene understanding?

We introduce a new large-scale dataset that addresses three core research problems in scene understanding: detecting non-iconic views (or non-canonical perspectives [12]) of objects, contextual reasoning between objects and the precise 2D localization of objects. For many categories of objects, there exists an iconic view. For example, when performing a web-based image search for the object category "bike," the top-ranked retrieved examples appear in profile, unobstructed near the center of a neatly composed photo. We posit that current recognition systems perform fairly well on iconic views, but struggle to recognize objects otherwise – in the

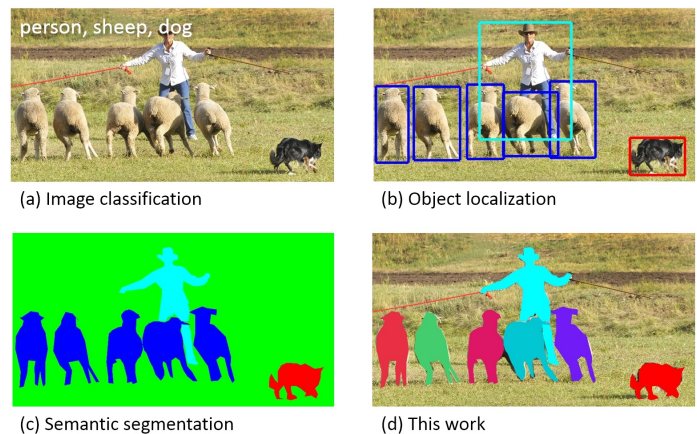


Fig. 1: While previous object recognition datasets have focused on (a) image classification, (b) object bounding box localization or (c) semantic pixel-level segmentation, we focus on (d) segmenting individual object instances. We introduce a large, richly-annotated dataset comprised of images depicting complex everyday scenes of common objects in their natural context.

background, partially occluded, amid clutter [13] – reflecting the composition of actual everyday scenes. We verify this experimentally; when evaluated on everyday scenes, models trained on our data perform better than those trained with prior datasets. A challenge is finding natural images that contain multiple objects. The identity of many objects can only be resolved using context, due to small size or ambiguous appearance in the image. To push research in contextual reasoning, images depicting scenes [3] rather than objects in isolation are necessary. Finally, we argue that detailed spatial understanding of object layout will be a core component of scene analysis. An object's spatial location can be defined coarsely using a bounding box [2] or with a precise pixel-level segmentation [14], [15], [16]. As we demonstrate, to measure either kind of localization performance it is essential for the dataset to have every instance of every object

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微软 COCO: 上下文中的常见物体

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摘要—我们提出了一个新的数据集，旨在通过将物体识别问题置于更广泛的场景理解背景下，推动物体识别技术的进步。该数据集通过收集包含常见物体于自然场景中的复杂日常场景图像来实现这一目标。我们采用逐实例分割的方式对物体进行标注，以辅助精确的物体定位。数据集中包含91类四岁儿童也能轻松识别的物体类型，在32.8万张图像中总计标注了250万个实例。我们通过新颖的用户界面发动大量众包工作者参与数据集的构建，实现了类别检测、实例定位和实例分割。我们详细分析了本数据集与PASCAL、ImageNet和SUN的统计特征对比。最后，我们使用可变形部件模型为边界框检测和分割检测结果提供了基线性能分析。

1 引言

计算机视觉的主要目标之一是理解视觉场景。场景理解涉及众多任务，包括识别存在的物体、在二维和三维空间中定位物体、确定物体及场景的属性、描述物体间的关系，并提供场景的语义描述。当前物体分类与检测数据集[1]、[2]、[3]、[4]帮助我们探索与场景理解相关的首要挑战。例如，包含空前数量图像的ImageNet数据集[1]近年来推动了物体分类与检测研究的突破性进展[5]、[6]、[7]。研究界还创建了包含物体属性[8]、场景属性[9]、关键点[10]及三维场景信息[11]的数据集。这引出了一个显而易见的问题：哪些数据集能最有效地推动我们朝着场景理解的终极目标继续前进？

我们引入了一个新的大规模数据集，旨在解决场景理解中的三个核心研究问题：检测物体的非标志性视角（或称非规范视角[12]）、物体间的上下文推理以及物体的精确二维定位。对于许多物体类别而言，存在一种标志性视角。例如，当在网络上搜索“自行车”这一物体类别时，排名靠前的检索结果通常呈现为侧面视角、无遮挡且位于构图整洁的照片中心。我们认为当前的识别系统在标志性视角上表现相当出色，但在其他情况下识别物体时则面临困难——例如

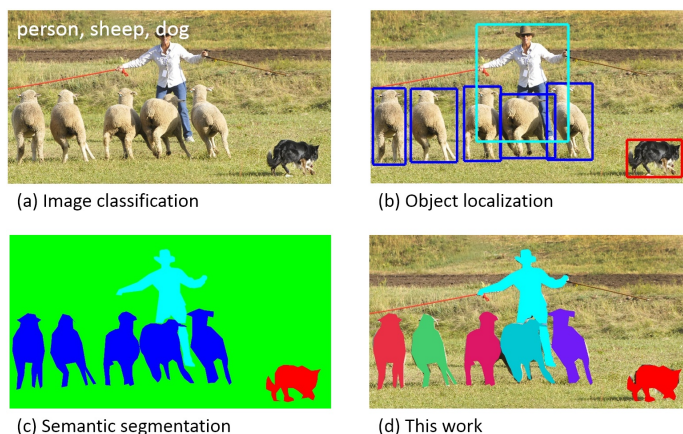


图1：以往的目标识别数据集主要关注（a）图像分类、（b）目标边界框定位或（c）语义像素级分割，而我们的研究重点在于（d）分割单个目标实例。我们引入了一个大规模、标注丰富的数据集，其中包含描绘常见物体在自然场景中复杂日常环境的图像。

背景、部分遮挡、处于杂乱之中[13]——这反映了实际日常场景的构成。我们通过实验验证了这一点；在日常场景中评估时，使用我们数据训练的模型表现优于使用先前数据集训练的模型。一个挑战在于寻找包含多个物体的自然图像。由于图像中物体尺寸过小或外观模糊，许多物体的身份只能通过上下文来解析。为了推动上下文推理的研究，需要描绘场景[3]而非孤立物体的图像。最后，我们认为对物体布局的详细空间理解将成为场景分析的核心组成部分。物体的空间位置可以通过边界框[2]进行粗略定义，或通过精确的像素级分割[14]、[15]、[16]来界定。正如我们所展示的，要衡量任何一种定位性能，数据集中必须包含每个物体的每个实例。

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category labeled and fully segmented. Our dataset is unique in its annotation of instance-level segmentation masks, Fig. 1.

To create a large-scale dataset that accomplishes these three goals we employed a novel pipeline for gathering data with extensive use of Amazon Mechanical Turk. First and most importantly, we harvested a large set of images containing contextual relationships and non-iconic object views. We accomplished this using a surprisingly simple yet effective technique that queries for pairs of objects in conjunction with images retrieved via scene-based queries [17], [3]. Next, each image was labeled as containing particular object categories using a hierarchical labeling approach [18]. For each category found, the individual instances were labeled, verified, and finally segmented. Given the inherent ambiguity of labeling, each of these stages has numerous tradeoffs that we explored in detail.

The Microsoft Common Objects in COntext (MS COCO) dataset contains 91 common object categories with 82 of them having more than 5,000 labeled instances, Fig. 6. In total the dataset has 2,500,000 labeled instances in 328,000 images. In contrast to the popular ImageNet dataset [1], COCO has fewer categories but more instances per category. This can aid in learning detailed object models capable of precise 2D localization. The dataset is also significantly larger in number of instances per category than the PASCAL VOC [2] and SUN [3] datasets. Additionally, a critical distinction between our dataset and others is the number of labeled instances per image which may aid in learning contextual information, Fig. 5. MS COCO contains considerably more object instances per image (7.7) as compared to ImageNet (3.0) and PASCAL (2.3). In contrast, the SUN dataset, which contains significant contextual information, has over 17 objects and “stuff” per image but considerably fewer object instances overall.

An abridged version of this work appeared in [19].

2 RELATED WORK

Throughout the history of computer vision research datasets have played a critical role. They not only provide a means to train and evaluate algorithms, they drive research in new and more challenging directions. The creation of ground truth stereo and optical flow datasets [20], [21] helped stimulate a flood of interest in these areas. The early evolution of object recognition datasets [22], [23], [24] facilitated the direct comparison of hundreds of image recognition algorithms while simultaneously pushing the field towards more complex problems. Recently, the ImageNet dataset [1] containing millions of images has enabled breakthroughs in both object classification and detection research using a new class of deep learning algorithms [5], [6], [7].

Datasets related to object recognition can be roughly split into three groups: those that primarily address object classification, object detection and semantic scene labeling. We address each in turn.

Image Classification The task of object classification requires binary labels indicating whether objects are present in an image; see Fig. 1(a). Early datasets of this type comprised images containing a single object with blank backgrounds, such as the MNIST handwritten digits [25] or COIL household objects [26]. Caltech 101 [22] and Caltech 256 [23] marked the transition to more realistic object images retrieved from the internet while also increasing the number of object categories to 101 and 256, respectively. Popular datasets in the machine learning community due to the larger number of training examples, CIFAR-10 and CIFAR-100 [27] offered 10 and 100 categories from a dataset of tiny 32×32 images [28]. While these datasets contained up to 60,000 images and hundreds of categories, they still only captured a small fraction of our visual world.

Recently, ImageNet [1] made a striking departure from the incremental increase in dataset sizes. They proposed the creation of a dataset containing 22k categories with 500-1000 images each. Unlike previous datasets containing entry-level categories [29], such as “dog” or “chair,” like [28], ImageNet used the WordNet Hierarchy [30] to obtain both entry-level and fine-grained [31] categories. Currently, the ImageNet dataset contains over 14 million labeled images and has enabled significant advances in image classification [5], [6], [7].

Object detection Detecting an object entails both stating that an object belonging to a specified class is present, and localizing it in the image. The location of an object is typically represented by a bounding box, Fig. 1(b). Early algorithms focused on face detection [32] using various ad hoc datasets. Later, more realistic and challenging face detection datasets were created [33]. Another popular challenge is the detection of pedestrians for which several datasets have been created [24], [4]. The Caltech Pedestrian Dataset [4] contains 350,000 labeled instances with bounding boxes.

For the detection of basic object categories, a multi-year effort from 2005 to 2012 was devoted to the creation and maintenance of a series of benchmark datasets that were widely adopted. The PASCAL VOC [2] datasets contained 20 object categories spread over 11,000 images. Over 27,000 object instance bounding boxes were labeled, of which almost 7,000 had detailed segmentations. Recently, a detection challenge has been created from 200 object categories using a subset of 400,000 images from ImageNet [34]. An impressive 350,000 objects have been labeled using bounding boxes.

Since the detection of many objects such as sunglasses, cellphones or chairs is highly dependent on contextual information, it is important that detection datasets contain objects in their natural environments. In our dataset we strive to collect images rich in contextual information. The use of bounding boxes also limits the accuracy for which detection algorithms may be evaluated. We propose the use of fully segmented instances to enable more accurate detector evaluation.

类别已标注且完全分割。我们的数据集在实例级分割掩码的标注方面独具特色，如图1所示。

为了创建一个能够实现这三个目标的大规模数据集，我们采用了一种新颖的数据收集流程，广泛利用了亚马逊土耳其机器人（Amazon Mechanical Turk）。首先也是最重要的一步，我们收集了大量包含上下文关系和非典型物体视角的图像。我们通过一种出奇简单却高效的技术实现了这一点，该技术通过查询物体对并结合基于场景查询检索到的图像来完成[17], [3]。接着，采用分层标注方法[18]对每张图像进行标注，标记出其中包含的特定物体类别。对于发现的每个类别，我们对单个实例进行标注、验证，并最终进行分割。考虑到标注固有的模糊性，每个阶段都存在许多权衡，我们对此进行了详细探讨。

微软上下文常见物体（MS COCO）数据集包含91个常见物体类别，其中82个类别拥有超过5,000个标注实例（图6）。该数据集总计在328,000张图像中包含2,500,000个标注实例。与流行的ImageNet数据集[1]相比，COCO的类别更少但每个类别的实例更多。这有助于学习能够精确定位二维位置的详细物体模型。该数据集每个类别的实例数量也显著超过PASCAL VOC[2]和SUN[3]数据集。此外，我们的数据集与其他数据集的关键区别在于每张图像的标注实例数量（这可能有助于学习上下文信息，见图5）。MS COCO每张图像包含的物体实例数（7.7个）明显多于ImageNet（3.0个）和PASCAL（2.3个）。相比之下，虽然SUN数据集包含丰富的上下文信息，每张图像有超过17个物体及“背景物”，但总体物体实例数量要少得多。

本作品的简略版曾发表于[19]。

2 相关工作

在计算机视觉研究的历史中，数据集始终扮演着关键角色。它们不仅为算法训练与评估提供基础，更推动研究朝着更具挑战性的新方向发展。立体视觉与光流真实数据集的建立[20][21]曾极大激发了这些领域的研究热潮。早期物体识别数据集[22][23][24]的演进，使数百种图像识别算法得以直接比较，同时推动该领域向更复杂的问题迈进。近年来，包含数百万图像的ImageNet数据集[1]通过新型深度学习算法[5][6][7]，在物体分类与检测研究中实现了突破性进展。

与物体识别相关的数据集大致可分为三类：主要针对物体分类、物体检测和语义场景标记的数据集。我们将逐一探讨这些类别。

图像分类 物体分类任务需要二元标签来指示图像中是否存在物体；见图1(a)。早期的此类数据集包含具有空白背景的单个物体图像，例如MNIST手写数字[25]或COIL家居物品[26]。Caltech 101[22]和Caltech 256[23]标志着向从互联网检索的更真实物体图像的过渡，同时将物体类别数量分别增加到101和256。由于训练样本数量更大而备受机器学习社区欢迎的CIFAR-10和CIFAR-100[27]，在微型 32×32 图像数据集[28]中提供了10个和100个类别。尽管这些数据集包含多达60,000张图像和数百个类别，但它们仍然只捕捉了我们视觉世界的一小部分。

最近，ImageNet [1] 在数据集规模的增长方式上做出了引人注目的突破。他们提出了一个包含22,000个类别、每个类别有500至1000张图像的数据集构建方案。与以往仅包含基础类别（如“狗”或“椅子”）的数据集（如[28]）不同，ImageNet利用WordNet层级结构[30]同时获取了基础类别和细粒度类别[31]。目前，ImageNet数据集已包含超过1400万张标注图像，并在图像分类领域[5]、[6]、[7]推动了显著进展。

目标检测 检测一个物体既需要声明存在属于特定类别的物体，又需要将其在图像中定位。物体的位置通常由边界框表示，如图1(b)所示。早期算法主要使用各种特定数据集进行人脸检测[32]。随后，出现了更真实且更具挑战性的人脸检测数据集[33]。另一个常见的挑战是行人检测，为此已创建了多个数据集[24]、[4]。Caltech行人数据集[4]包含350,000个带有边界框的标注实例。

对于基础物体类别的检测，从2005年到2012年，一项持续多年的努力致力于创建和维护一系列被广泛采用的基准数据集。PASCAL VOC [2] 数据集包含了分布在11,000张图像中的20个物体类别。超过27,000个物体实例的边界框被标注，其中近7,000个具有详细的分割标注。最近，基于ImageNet [34] 的400,000张图像子集，一个包含200个物体类别的检测挑战赛被创建。令人印象深刻的是，已有350,000个物体通过边界框完成了标注。

由于许多物体（如太阳镜、手机或椅子）的检测高度依赖于上下文信息，因此检测数据集包含自然环境中的物体至关重要。在我们的数据集中，我们致力于收集富含上下文信息的图像。边界框的使用也限制了检测算法评估的准确性。我们建议使用完全分割的实例来实现更精确的检测器评估。



Fig. 2: Example of (a) iconic object images, (b) iconic scene images, and (c) non-iconic images.

Semantic scene labeling The task of labeling semantic objects in a scene requires that each pixel of an image be labeled as belonging to a category, such as sky, chair, floor, street, etc. In contrast to the detection task, individual instances of objects do not need to be segmented, Fig. 1(c). This enables the labeling of objects for which individual instances are hard to define, such as grass, streets, or walls. Datasets exist for both indoor [11] and outdoor [35], [14] scenes. Some datasets also include depth information [11]. Similar to semantic scene labeling, our goal is to measure the pixel-wise accuracy of object labels. However, we also aim to distinguish between individual instances of an object, which requires a solid understanding of each object’s extent.

A novel dataset that combines many of the properties of both object detection and semantic scene labeling datasets is the SUN dataset [3] for scene understanding. SUN contains 908 scene categories from the WordNet dictionary [30] with segmented objects. The 3,819 object categories span those common to object detection datasets (person, chair, car) and to semantic scene labeling (wall, sky, floor). Since the dataset was collected by finding images depicting various scene types, the number of instances per object category exhibits the long tail phenomenon. That is, a few categories have a large number of instances (wall: 20,213, window: 16,080, chair: 7,971) while most have a relatively modest number of instances (boat: 349, airplane: 179, floor lamp: 276). In our dataset, we ensure that each object category has a significant number of instances, Fig. 5.

Other vision datasets Datasets have spurred the advancement of numerous fields in computer vision. Some notable datasets include the Middlebury datasets for stereo vision [20], multi-view stereo [36] and optical flow [21]. The Berkeley Segmentation Data Set (BSDS500) [37] has been used extensively to evaluate both segmentation and edge detection algorithms. Datasets have also been created to recognize both scene [9] and object attributes [8], [38]. Indeed, numerous areas of vision have benefited from challenging datasets that helped catalyze progress.

3 IMAGE COLLECTION

We next describe how the object categories and candidate images are selected.

3.1 Common Object Categories

The selection of object categories is a non-trivial exercise. The categories must form a representative set of all categories, be relevant to practical applications and occur with high enough frequency to enable the collection of a large dataset. Other important decisions are whether to include both “thing” and “stuff” categories [39] and whether fine-grained [31], [1] and object-part categories should be included. “Thing” categories include objects for which individual instances may be easily labeled (person, chair, car) where “stuff” categories include materials and objects with no clear boundaries (sky, street, grass). Since we are primarily interested in precise localization of object instances, we decided to only include “thing” categories and not “stuff.” However, since “stuff” categories can provide significant contextual information, we believe the future labeling of “stuff” categories would be beneficial.

The specificity of object categories can vary significantly. For instance, a dog could be a member of the “mammal”, “dog”, or “German shepherd” categories. To enable the practical collection of a significant number of instances per category, we chose to limit our dataset to entry-level categories, i.e. category labels that are commonly used by humans when describing objects (dog, chair, person). It is also possible that some object categories may be parts of other object categories. For instance, a face may be part of a person. We anticipate the inclusion of object-part categories (face, hands, wheels) would be beneficial for many real-world applications.

We used several sources to collect entry-level object categories of “things.” We first compiled a list of categories by combining categories from PASCAL VOC [2] and a subset of the 1200 most frequently used words that denote visually identifiable objects [40]. To further augment our set of candidate categories, several children ranging in ages from 4 to 8 were asked to name every



(a) Iconic object images

(b) Iconic scene images

(c) Non-iconic images

图2: (a)标志性物体图像、(b)标志性场景图像和(c)非标志性图像的示例。

语义场景标注 为场景中的语义对象进行标注的任务，要求将图像的每个像素标记为属于某个类别，例如天空、椅子、地板、街道等。与检测任务不同，该任务无需分割对象的独立实例，如图1(c)所示。这使得那些难以定义独立实例的对象（如草地、街道或墙壁）也能被标注。现有数据集涵盖室内[11]和室外[35][14]场景，部分数据集还包含深度信息[11]。与语义场景标注类似，我们的目标是衡量对象标签的像素级准确性。但同时，我们也致力于区分对象的各个独立实例，这需要对每个对象的范围有清晰的理解。

一种结合了目标检测和语义场景标注数据集诸多特性的新颖数据集是用于场景理解的SUN数据集[3]。SUN包含来自WordNet词典[30]的908个场景类别，并带有分割好的物体。其3,819个物体类别涵盖了目标检测数据集常见类别（人、椅子、汽车）和语义场景标注常见类别（墙壁、天空、地板）。由于该数据集通过收集描绘各类场景的图像构建，每个物体类别的实例数量呈现出长尾现象。即少数类别拥有大量实例（墙壁：20,213个，窗户：16,080个，椅子：7,971个），而大多数类别实例数量相对有限（船只：349个，飞机：179个，落地灯：276个）。在我们的数据集中，我们确保每个物体类别都拥有足够数量的实例，如图5所示。

其他视觉数据集数据集推动了计算机视觉众多领域的发展。一些著名的数据集包括用于立体视觉的Middlebury数据集[20]、多视图立体视觉[36]和光流[21]。伯克利分割数据集（BSDS500）[37]已被广泛用于评估分割和边缘检测算法。此外，还创建了用于识别场景[9]和物体属性[8]、[38]的数据集。事实上，许多视觉领域都受益于具有挑战性的数据集，这些数据集有助于催化研究进展。

3 图像采集

接下来，我们将描述如何选择对象类别和候选图像。

3.1 常见对象类别

对象类别的选择并非易事。这些类别必须构成所有类别的代表性集合，与实际应用相关，并且出现频率足够高，以便能够收集大型数据集。其他重要决策包括是否同时包含“物体”和“材质”类别[39]，以及是否纳入细粒度类别[31][1]和对象部件类别。“物体”类别包含易于标记独立实例的对象（人、椅子、汽车），而“材质”类别则涵盖没有明确边界的材料和对象（天空、街道、草地）。由于我们主要关注对象实例的精确定位，因此决定仅包含“物体”类别，不纳入“材质”类别。然而，鉴于“材质”类别能提供重要的上下文信息，我们认为未来对“材质”类别的标注将大有裨益。

物体类别的具体程度可以有很大差异。例如，一只狗可能属于“哺乳动物”“狗”或“德国牧羊犬”等不同层级的类别。为了实际收集每个类别的大量实例，我们选择将数据集限制在基础层级类别，即人类描述物体时常用的类别标签（狗、椅子、人）。某些物体类别也可能是其他物体类别的组成部分，例如人脸可能是人体的一部分。我们预计包含物体部件类别（面部、手部、车轮）将对许多实际应用有所裨益。

我们利用多种来源收集了“物品”的入门级物体类别。首先，我们通过整合PASCAL VOC [2] 的类别以及表示视觉可识别物体的1200个最常用词子集 [40]，初步编制了一个类别列表。为了进一步扩充候选类别集，我们邀请了几名年龄在4至8岁之间的儿童，请他们逐一命名

Annotation Pipeline

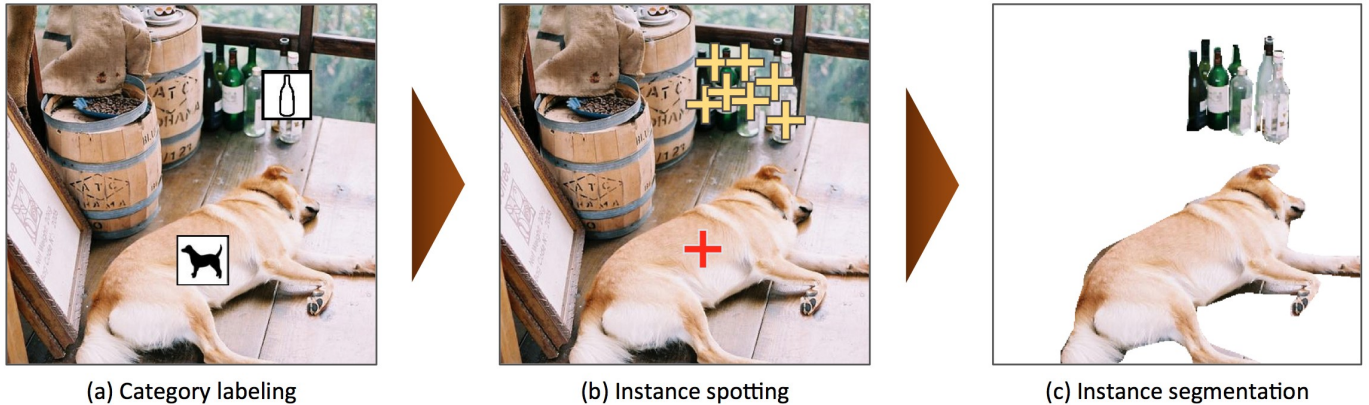


Fig. 3: Our annotation pipeline is split into 3 primary tasks: (a) labeling the categories present in the image (§4.1), (b) locating and marking all instances of the labeled categories (§4.2), and (c) segmenting each object instance (§4.3).

object they see in indoor and outdoor environments. The final 272 candidates may be found in the appendix. Finally, the co-authors voted on a 1 to 5 scale for each category taking into account how commonly they occur, their usefulness for practical applications, and their diversity relative to other categories. The final selection of categories attempts to pick categories with high votes, while keeping the number of categories per super-category (animals, vehicles, furniture, etc.) balanced. Categories for which obtaining a large number of instances (greater than 5,000) was difficult were also removed. To ensure backwards compatibility all categories from PASCAL VOC [2] are also included. Our final list of 91 proposed categories is in Fig. 5(a).

3.2 Non-iconic Image Collection

Given the list of object categories, our next goal was to collect a set of candidate images. We may roughly group images into three types, Fig. 2: iconic-object images [41], iconic-scene images [3] and non-iconic images. Typical iconic-object images have a single large object in a canonical perspective centered in the image, Fig. 2(a). Iconic-scene images are shot from canonical viewpoints and commonly lack people, Fig. 2(b). Iconic images have the benefit that they may be easily found by directly searching for specific categories using Google or Bing image search. While iconic images generally provide high quality object instances, they can lack important contextual information and non-canonical viewpoints.

Our goal was to collect a dataset such that a majority of images are non-iconic, Fig. 2(c). It has been shown that datasets containing more non-iconic images are better at generalizing [42]. We collected non-iconic images using two strategies. First as popularized by PASCAL VOC [2], we collected images from Flickr which tends to have fewer iconic images. Flickr contains photos uploaded by amateur photographers with searchable metadata and keywords. Second, we did not search for object categories in isolation. A search for “dog” will tend to return

iconic images of large, centered dogs. However, if we searched for pairwise combinations of object categories, such as “dog + car” we found many more non-iconic images. Surprisingly, these images typically do not just contain the two categories specified in the search, but numerous other categories as well. To further supplement our dataset we also searched for scene/object category pairs, see the appendix. We downloaded at most 5 photos taken by a single photographer within a short time window. In the rare cases in which enough images could not be found, we searched for single categories and performed an explicit filtering stage to remove iconic images. The result is a collection of 328,000 images with rich contextual relationships between objects as shown in Figs. 2(c) and 6.

4 IMAGE ANNOTATION

We next describe how we annotated our image collection. Due to our desire to label over 2.5 million object instances, the design of a cost efficient yet high quality annotation pipeline was critical. The annotation pipeline is outlined in Fig. 3. For all crowdsourcing tasks we used workers on Amazon’s Mechanical Turk (AMT). Our user interfaces are described in detail in the appendix. Note that, since the original version of this work [19], we have taken a number of steps to further improve the quality of the annotations. In particular, we have increased the number of annotators for the category labeling and instance spotting stages to eight. We also added a stage to verify the instance segmentations.

4.1 Category Labeling

The first task in annotating our dataset is determining which object categories are present in each image, Fig. 3(a). Since we have 91 categories and a large number of images, asking workers to answer 91 binary classification questions per image would be prohibitively expensive. Instead, we used a hierarchical approach [18].

Annotation Pipeline

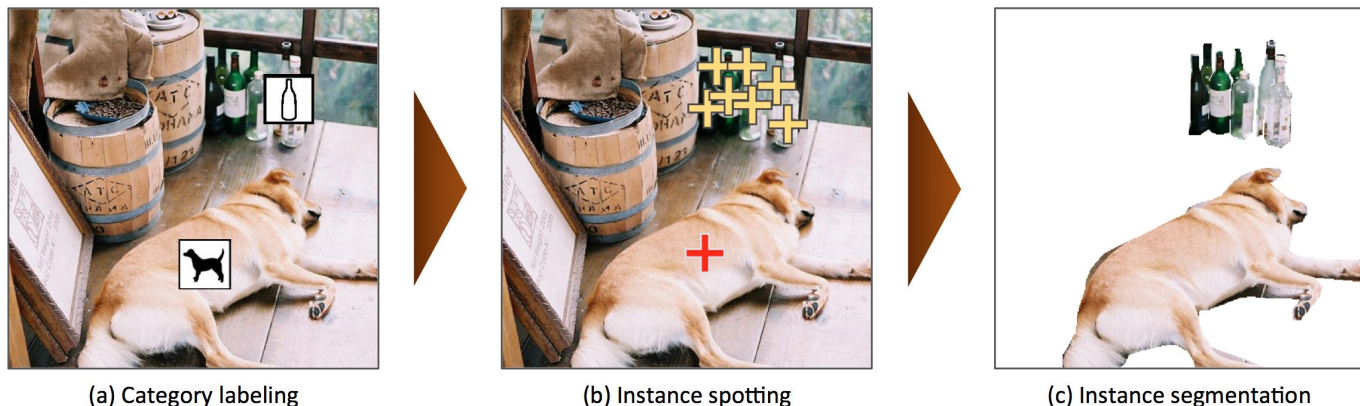


图3: 我们的标注流程分为三个主要任务: (a) 标注图像中存在的类别 (§4.1), (b) 定位并标记所有已标注类别的实例 (§4.2), 以及(c) 分割每个对象实例 (§4.3)。

他们在室内和室外环境中看到的物体。最终272个候选类别可在附录中找到。最后, 合著者根据每个类别的常见程度、实际应用价值以及与其他类别的差异度, 以1到5分进行投票。最终类别的筛选旨在选择高票类别, 同时保持每个超类别(动物、车辆、家具等)下的数量均衡。对于难以获取大量实例(超过5,000个)的类别也予以剔除。为确保向后兼容性, PASCAL VOC [2]的所有类别均被保留。图5(a)展示了我们最终提出的91个类别列表。

标志性图像通常是那些大型、居中的狗。然而, 如果我们搜索物体类别的成对组合, 比如“狗+汽车”, 我们会发现更多非标志性的图像。令人惊讶的是, 这些图像通常不仅包含搜索中指定的两个类别, 还有许多其他类别。为了进一步补充我们的数据集, 我们还搜索了场景/物体类别对, 详见附录。我们最多下载了同一摄影师在短时间内拍摄的5张照片。在极少数情况下, 如果找不到足够的图像, 我们会搜索单一类别, 并进行明确的筛选阶段以移除标志性图像。最终, 我们收集了328,000张图像, 这些图像中物体之间具有丰富的上下文关系, 如图2(c)和图6所示。

3.2 非图标图像收集

给定物体类别列表后, 我们的下一个目标是收集一组候选图像。我们可以将图像大致分为三种类型, 如图2所示: 标志性物体图像[41]、标志性场景图像[3]和非标志性图像。典型的标志性物体图像通常以规范视角呈现单个大型物体并居于画面, 如图2(a)所示。标志性场景图像则从规范视角拍摄, 通常不包含人物, 如图2(b)所示。标志性图像的优势在于, 通过谷歌或必应图片搜索直接检索特定类别即可轻松获取。虽然标志性图像通常能提供高质量的物体实例, 但它们可能缺乏重要的上下文信息和非规范视角。

我们的目标是收集一个数据集, 使得大多数图像都是非标志性的, 如图2(c)所示。研究表明, 包含更多非标志性图像的数据集在泛化能力上表现更佳[42]。我们采用两种策略收集非标志性图像。首先, 借鉴PASCAL VOC[2]的流行做法, 我们从Flickr收集图像, 该平台上的标志性图像通常较少。Flickr包含业余摄影师上传的照片, 并带有可搜索的元数据和关键词。其次, 我们并未单独搜索物体类别。例如, 搜索“狗”往往会返回

4 图像标注

接下来, 我们将描述如何为我们的图像集进行标注。由于我们需要标注超过250万个物体实例, 设计一个成本效益高且质量优良的标注流程至关重要。图3概述了该标注流程。在所有众包任务中, 我们使用了亚马逊土耳其机器人(AMT)上的工作者。附录中详细描述了我们的用户界面。请注意, 自本工作的原始版本[19]以来, 我们已采取多项措施进一步提高标注质量。具体来说, 我们将类别标注和实例识别阶段的标注人员数量增加到了八人。我们还增加了一个阶段来验证实例分割结果。

4.1 类别标注

为我们的数据集进行标注的首要任务是确定每张图像中存在哪些物体类别, 如图3(a)所示。由于我们拥有91个类别和大量图像, 若要求标注人员为每张图像回答91个二元分类问题将成本过高。因此, 我们采用了分层方法[18]。

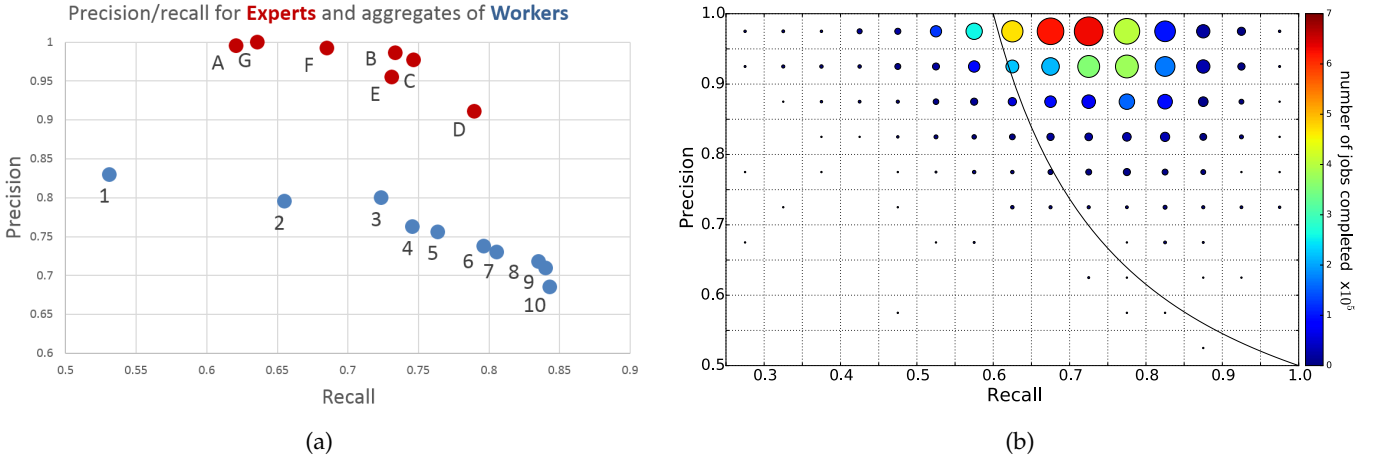


Fig. 4: Worker precision and recall for the category labeling task. (a) The union of multiple AMT workers (blue) has better recall than any expert (red). Ground truth was computed using majority vote of the experts. (b) Shows the number of workers (circle size) and average number of jobs per worker (circle color) for each precision/recall range. Most workers have high precision; such workers generally also complete more jobs. For this plot ground truth for each worker is the *union* of responses from all other AMT workers. See §4.4 for details.

We group the object categories into 11 super-categories (see the appendix). For a given image, a worker was presented with each group of categories in turn and asked to indicate whether any instances exist for that super-category. This greatly reduces the time needed to classify the various categories. For example, a worker may easily determine no animals are present in the image without having to specifically look for cats, dogs, etc. If a worker determines instances from the super-category (animal) are present, for each subordinate category (dog, cat, etc.) present, the worker must drag the category’s icon onto the image over one instance of the category. The placement of these icons is critical for the following stage. We emphasize that only a single instance of each category needs to be annotated in this stage. To ensure high recall, 8 workers were asked to label each image. A category is considered present if any worker indicated the category; false positives are handled in subsequent stages. A detailed analysis of performance is presented in §4.4. This stage took $\sim 20k$ worker hours to complete.

4.2 Instance Spotting

In the next stage all instances of the object categories in an image were labeled, Fig. 3(b). In the previous stage each worker labeled one instance of a category, but multiple object instances may exist. Therefore, for each image, a worker was asked to place a cross on top of each instance of a specific category found in the previous stage. To boost recall, the location of the instance found by a worker in the previous stage was shown to the current worker. Such priming helped workers quickly find an initial instance upon first seeing the image. The workers could also use a magnifying glass to find small instances. Each worker was asked to label at most 10 instances of a given category per image. Each image was labeled by 8 workers for a total of $\sim 10k$ worker hours.

4.3 Instance Segmentation

Our final stage is the laborious task of segmenting each object instance, Fig. 3(c). For this stage we modified the excellent user interface developed by Bell et al. [16] for image segmentation. Our interface asks the worker to segment an object instance specified by a worker in the previous stage. If other instances have already been segmented in the image, those segmentations are shown to the worker. A worker may also indicate there are no object instances of the given category in the image (implying a false positive label from the previous stage) or that all object instances are already segmented.

Segmenting 2,500,000 object instances is an extremely time consuming task requiring over 22 worker hours per 1,000 segmentations. To minimize cost we only had a single worker segment each instance. However, when first completing the task, most workers produced only coarse instance outlines. As a consequence, we required all workers to complete a training task for each object category. The training task required workers to segment an object instance. Workers could not complete the task until their segmentation adequately matched the ground truth. The use of a training task vastly improved the quality of the workers (approximately 1 in 3 workers passed the training stage) and resulting segmentations. Example segmentations may be viewed in Fig. 6.

While the training task filtered out most bad workers, we also performed an explicit verification step on each segmented instance to ensure good quality. Multiple workers (3 to 5) were asked to judge each segmentation and indicate whether it matched the instance well or not. Segmentations of insufficient quality were discarded and the corresponding instances added back to the pool of unsegmented objects. Finally, some approved workers consistently produced poor segmentations; all work obtained from such workers was discarded.

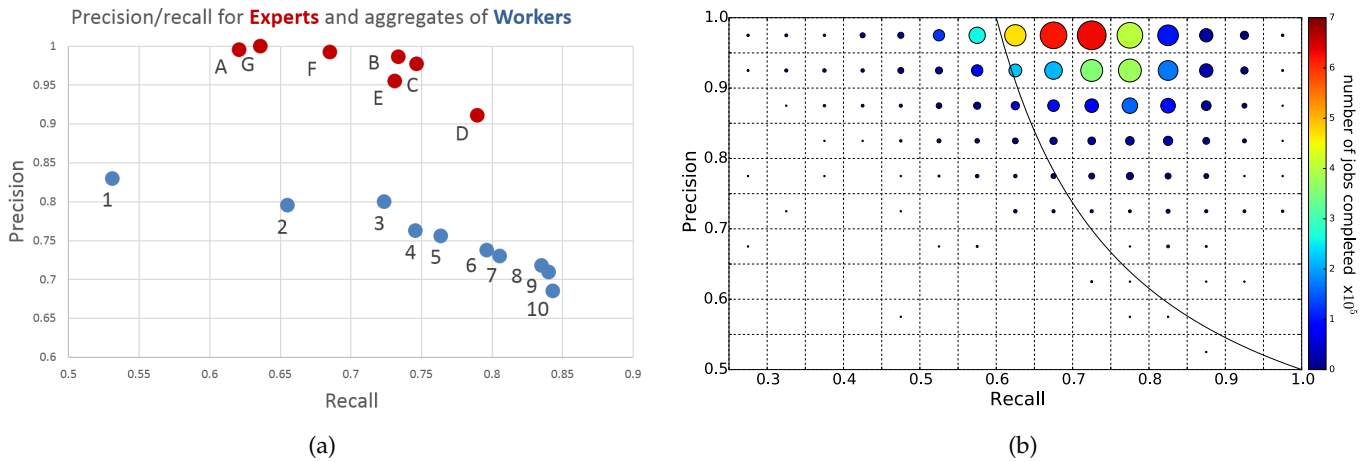


图4：类别标注任务中工人的精确率与召回率。(a) 多位AMT工人的结果并集（蓝色）比任何专家（红色）具有更好的召回率。真实值通过专家多数投票计算得出。(b) 展示了每个精确率/召回率区间对应的工人数量（圆圈大小）和工人平均任务数（圆圈颜色）。大多数工人具有高精确率；这类工人通常也完成了更多任务。本图中每位工人的真实值采用所有其他AMT工人回复的union。详见§4.4节。

我们将对象类别分为11个超类别（详见附录）。对于给定图像，工作人员会依次查看每组类别，并判断该超类别是否存在任何实例。这大大减少了分类各类别所需的时间。例如，工作人员可以轻松判断图像中不存在动物，而无需专门寻找猫、狗等具体类别。若工作人员确定存在某超类别（如动物）的实例，则需为该超类别下每个存在的子类别（如狗、猫等）将其图标拖拽至图像中该类别的**一个实例**上。这些图标的放置位置对后续阶段至关重要。需要强调的是，此阶段每个类别仅需标注**单个实例**。为确保高召回率，每张图像由8名工作人员进行标注。只要有任何工作人员标注了某类别，即判定该类别存在；误报问题将在后续阶段处理。具体性能分析详见§4.4节。此阶段总计耗时~2万工时。

4.2 实例识别

在下一阶段，图像中所有物体类别的实例均被标注，如图3(b)所示。前一阶段每位标注员仅标注一个类别的单个实例，但实际可能存在多个物体实例。因此，针对每张图像，我们要求标注员在前一阶段已识别出的特定类别所有实例上方放置十字标记。为提升召回率，系统会向当前标注员展示前一阶段其他标注员已标注的实例位置。这种提示机制能帮助标注员在初次查看图像时快速定位初始实例。标注员还可使用放大镜工具查找微小实例。每张图像中，每位标注员被要求对给定类别最多标注10个实例。每张图像由8位标注员共同处理，总计耗费~万标注工时。

4.3 实例分割

我们的最后阶段是分割每个对象实例的繁琐任务，如图3(c)所示。在此阶段，我们改进了Bell等人[16]开发的优秀图像分割用户界面。我们的界面要求工作人员分割由前一阶段工作人员指定的对象实例。如果图像中已有其他实例被分割，这些分割结果会展示给工作人员。工作人员也可以指出图像中没有给定类别的对象实例（这意味着前一阶段的标签是误报），或者所有对象实例都已被分割。

分割250万个对象实例是一项极其耗时的任务，每完成1000个分割需要超过22个工时。为降低成本，我们仅安排单名工人处理每个实例。然而，在初期完成任务时，大多数工人仅能勾勒出粗略的实例轮廓。因此，我们要求所有工人针对每个对象类别完成训练任务。该训练任务要求工人对一个对象实例进行分割，只有当其分割结果与真实标注充分匹配时，才能通过任务。训练任务的实施显著提升了工人的作业质量（约三分之一的工人通过训练阶段），并改善了最终的分割效果。示例分割结果可参见图6。

尽管训练任务过滤掉了大部分表现不佳的工作人员，我们还对每个分割实例进行了明确的验证步骤，以确保高质量。我们邀请了多位工作人员（3至5名）对每个分割进行评判，并指出其是否与实例匹配良好。质量不足的分割被丢弃，相应的实例被重新加入未分割对象池中。最后，一些已获批准的工作人员持续产出低质量分割；从这些工作人员处获得的所有工作成果均被废弃。

For images containing 10 object instances or fewer of a given category, every instance was individually segmented (note that in some images up to 15 instances were segmented). Occasionally the number of instances is drastically higher; for example, consider a dense crowd of people or a truckload of bananas. In such cases, many instances of the same category may be tightly grouped together and distinguishing individual instances is difficult. After 10-15 instances of a category were segmented in an image, the remaining instances were marked as “crowds” using a single (possibly multi-part) segment. For the purpose of evaluation, areas marked as crowds will be ignored and not affect a detector’s score. Details are given in the appendix.

4.4 Annotation Performance Analysis

We analyzed crowd worker quality on the category labeling task by comparing to dedicated expert workers, see Fig. 4(a). We compared precision and recall of seven expert workers (co-authors of the paper) with the results obtained by taking the union of one to ten AMT workers. Ground truth was computed using majority vote of the experts. For this task recall is of primary importance as false positives could be removed in later stages. Fig. 4(a) shows that the union of 8 AMT workers, the same number as was used to collect our labels, achieved greater recall than any of the expert workers. Note that worker recall saturates at around 9-10 AMT workers.

Object category presence is often ambiguous. Indeed as Fig. 4(a) indicates, even dedicated experts often disagree on object presence, e.g. due to inherent ambiguity in the image or disagreement about category definitions. For any unambiguous examples having a probability of over 50% of being annotated, the probability all 8 annotators missing such a case is at most $.5^8 \approx .004$. Additionally, by observing how recall increased as we added annotators, we estimate that in practice over 99% of all object categories not later rejected as false positives are detected given 8 annotators. Note that a similar analysis may be done for instance spotting in which 8 annotators were also used.

Finally, Fig. 4(b) re-examines precision and recall of AMT workers on category labeling on a much larger set of images. The number of workers (circle size) and average number of jobs per worker (circle color) is shown for each precision/recall range. Unlike in Fig. 4(a), we used a leave-one-out evaluation procedure where a category was considered present if *any* of the remaining workers named the category. Therefore, overall worker precision is substantially higher. Workers who completed the most jobs also have the highest precision; all jobs from workers below the black line were rejected.

4.5 Caption Annotation

We added five written caption descriptions to each image in MS COCO. A full description of the caption statistics and how they were gathered will be provided shortly in a separate publication.

5 DATASET STATISTICS

Next, we analyze the properties of the Microsoft Common Objects in COntext (MS COCO) dataset in comparison to several other popular datasets. These include ImageNet [1], PASCAL VOC 2012 [2], and SUN [3]. Each of these datasets varies significantly in size, list of labeled categories and types of images. ImageNet was created to capture a large number of object categories, many of which are fine-grained. SUN focuses on labeling scene types and the objects that commonly occur in them. Finally, PASCAL VOC’s primary application is object detection in natural images. MS COCO is designed for the detection and segmentation of objects occurring in their natural context.

The number of instances per category for all 91 categories is shown in Fig. 5(a). A summary of the datasets showing the number of object categories and the number of instances per category is shown in Fig. 5(d). While MS COCO has fewer categories than ImageNet and SUN, it has more instances per category which we hypothesize will be useful for learning complex models capable of precise localization. In comparison to PASCAL VOC, MS COCO has both more categories and instances.

An important property of our dataset is we strive to find non-iconic images containing objects in their natural context. The amount of contextual information present in an image can be estimated by examining the average number of object categories and instances per image, Fig. 5(b, c). For ImageNet we plot the object detection validation set, since the training data only has a single object labeled. On average our dataset contains 3.5 categories and 7.7 instances per image. In comparison ImageNet and PASCAL VOC both have less than 2 categories and 3 instances per image on average. Another interesting observation is only 10% of the images in MS COCO have only one category per image, in comparison, over 60% of images contain a single object category in ImageNet and PASCAL VOC. As expected, the SUN dataset has the most contextual information since it is scene-based and uses an unrestricted set of categories.

Finally, we analyze the average size of objects in the datasets. Generally smaller objects are harder to recognize and require more contextual reasoning to recognize. As shown in Fig. 5(e), the average sizes of objects is smaller for both MS COCO and SUN.

6 DATASET SPLITS

To accommodate a faster release schedule, we split the MS COCO dataset into two roughly equal parts. The first half of the dataset was released in 2014, the second half will be released in 2015. The 2014 release contains 82,783 training, 40,504 validation, and 40,775 testing images (approximately $\frac{1}{2}$ train, $\frac{1}{4}$ val, and $\frac{1}{4}$ test). There are nearly 270k segmented people and a total of 886k segmented object instances in the 2014 train+val data alone. The cumulative 2015 release will contain a total of 165,482 train, 81,208 val, and 81,434 test images.

对于包含给定类别10个或更少物体实例的图像，每个实例都进行了单独分割（请注意，在某些图像中分割了多达15个实例）。偶尔实例数量会显著增加；例如，考虑密集的人群或一卡车香蕉。在这种情况下，同一类别的许多实例可能紧密聚集在一起，区分单个实例很困难。当图像中某个类别已分割10-15个实例后，剩余实例将使用单个（可能是多部分的）分割标记为“人群”。为了评估目的，标记为人群的区域将被忽略，不会影响检测器的得分。详细信息见附录。

4.4 标注性能分析

我们通过比较众包工作者与专业专家工作者在类别标注任务上的表现来分析众包工作者的质量，具体见图4(a)。我们将七位专家工作者（论文合著者）的精确率和召回率与一到十位AMT工作者结果取并集后得到的结果进行了比较。专家多数投票的结果被用作基准真值。在此任务中，召回率至关重要，因为误报可以在后续阶段被排除。图4(a)显示，8位AMT工作者（与我们收集标签时所用数量相同）的并集所达到的召回率高于任何一位专家工作者。值得注意的是，工作者的召回率在约9-10位AMT工作者时趋于饱和。

物体类别的存在性常常是模糊的。确实，如图4(a)所示，即使是专业专家也常对物体是否存在意见不一，例如由于图像本身固有的模糊性或对类别定义的分歧。对于任何标注概率超过50%的明确样本，所有8位标注者均遗漏该情况的概率最多为 $5^8 \approx .004$ 。此外，通过观察增加标注者数量时召回率的提升情况，我们估计在实际操作中，给定8位标注者时，超过99%的物体类别（后续未被判定为误报）均能被检测到。需要注意的是，类似分析也可应用于同样使用8位标注者的实例识别任务中。

最后，图4(b)在一个更大的图像集上重新检验了AMT工作者在类别标注上的精确率和召回率。图中每个精确率/召回率区间都显示了工作者数量（圆圈大小）和每位工作者的平均任务量（圆圈颜色）。与图4(a)不同，我们采用了留一法评估流程：若剩余工作者中有*any*的人标注了该类别，则判定该类别存在。因此，整体工作者的精确率显著提高。完成任务量最多的工作者同时具有最高的精确率；所有位于黑线以下的工作者提交的任务均被拒绝。

4.5 字幕标注

我们在MS COCO数据集的每张图像上添加了三条书面描述。关于这些描述的统计信息及其收集方式的完整说明，将在不久后通过另一篇出版物单独提供。

5 数据集统计

接下来，我们分析微软上下文常见对象（MS COCO）数据集相较于其他几个流行数据集的特性。这些数据集包括ImageNet[1]、PASCAL VOC 2012[2]和SUN[3]。这些数据集在规模、标注类别列表和图像类型方面均存在显著差异。ImageNet的创建旨在涵盖大量物体类别，其中许多属于细粒度分类。SUN侧重于标注场景类型及其常见物体。最后，PASCAL VOC主要应用于自然图像中的物体检测。MS COCO则专为自然语境中物体的检测与分割而设计。

所有91个类别的每个类别的实例数量如图5(a)所示。数据集摘要显示了对象类别数量及每个类别的实例数量，如图5(d)所示。虽然MS COCO的类别数量少于ImageNet和SUN，但其每个类别的实例更多，我们推测这将有助于学习能够精确定位的复杂模型。与PASCAL VOC相比，MS COCO在类别数量和实例数量上都更占优势。

我们数据集的一个重要特性是，我们致力于寻找包含自然场景中物体的非标志性图像。图像中存在的上下文信息量可以通过检查每张图像的平均物体类别数和实例数来估算，如图5(b, c)所示。对于ImageNet，我们绘制了物体检测验证集的数据，因为其训练数据仅标注了单个物体。平均而言，我们的数据集每张图像包含3.5个类别和7.7个实例。相比之下，ImageNet和PASCAL VOC平均每张图像都少于2个类别和3个实例。另一个有趣的观察是，MS COCO中只有10%的图像每张仅含一个类别，而ImageNet和PASCAL VOC中超过60%的图像仅包含单一物体类别。正如预期的那样，SUN数据集拥有最多的上下文信息，因为它是基于场景的，并且使用了无限制的类别集合。

最后，我们分析了数据集中物体的平均尺寸。通常，较小的物体更难识别，需要更多的上下文推理才能辨认。如图5(e)所示，MS COCO和SUN数据集中物体的平均尺寸都较小。

6 数据集划分

为了适应更快的发布节奏，我们将MS COCO数据集大致平均分为两部分。数据集的前半部分于2014年发布，后半部分将于2015年发布。2014年发布的数据包含82,783张训练图像、40,504张验证图像和40,775张测试图像（约 $\frac{1}{2}$ 训练、 $\frac{1}{4}$ 验证和 $\frac{1}{4}$ 测试）。仅2014年的训练+验证数据中就包含近27万个分割人物和总计88.6万个分割目标实例。2015年累计发布的数据将总共包含165,482张训练图像、81,208张验证图像和81,434张测试图像。

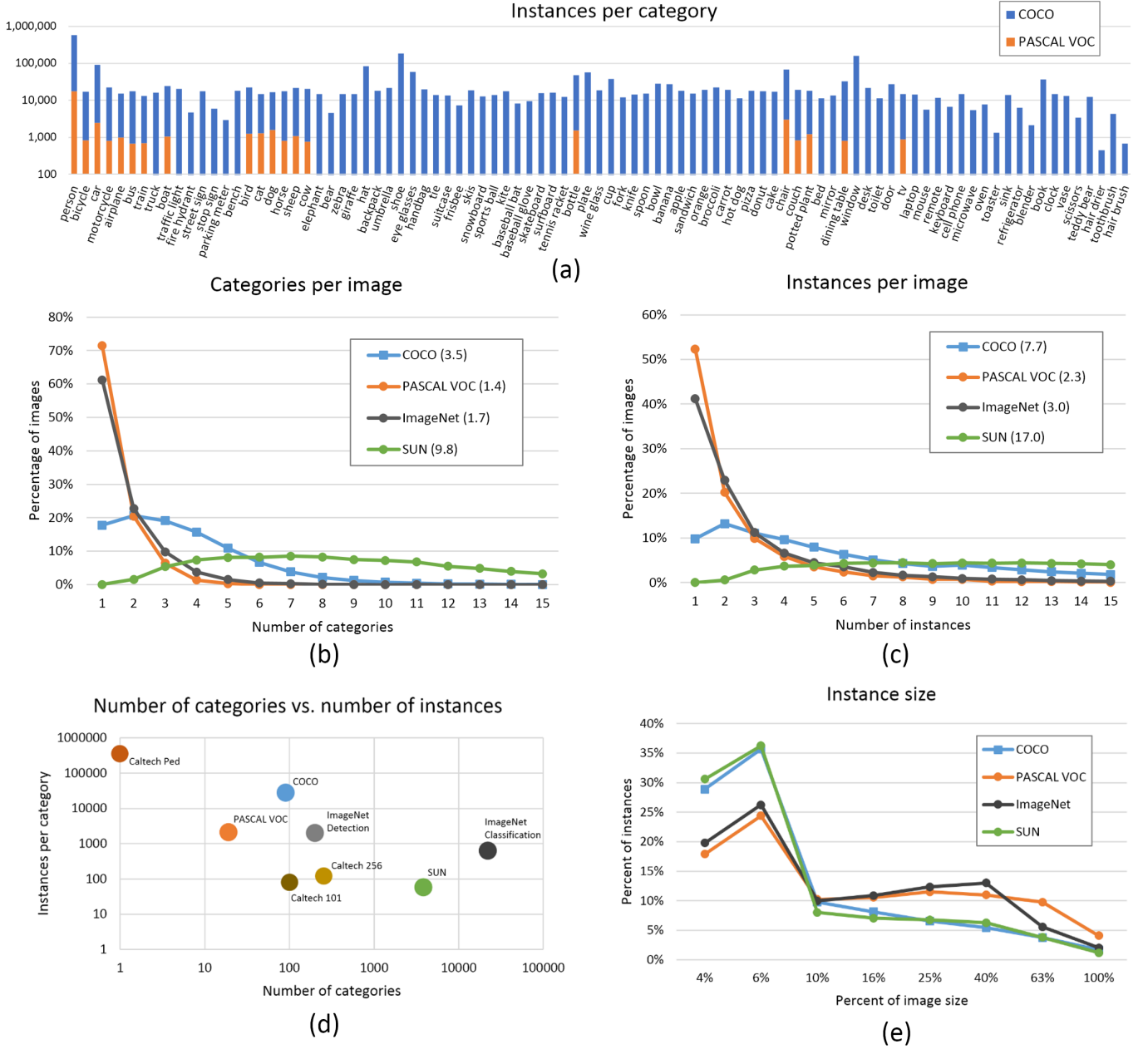


Fig. 5: (a) Number of annotated instances per category for MS COCO and PASCAL VOC. (b,c) Number of annotated categories and annotated instances, respectively, per image for MS COCO, ImageNet Detection, PASCAL VOC and SUN (average number of categories and instances are shown in parentheses). (d) Number of categories vs. the number of instances per category for a number of popular object recognition datasets. (e) The distribution of instance sizes for the MS COCO, ImageNet Detection, PASCAL VOC and SUN datasets.

We took care to minimize the chance of near-duplicate images existing across splits by explicitly removing near duplicates (detected with [43]) and grouping images by photographer and date taken.

Following established protocol, annotations for train and validation data will be released, but not for test. We are currently finalizing the evaluation server for automatic evaluation on the test set. A full discussion of evaluation metrics will be added once the evaluation

server is complete.

Note that we have limited the 2014 release to a subset of 80 categories. We did not collect segmentations for the following 11 categories: hat, shoe, eyeglasses (too many instances), mirror, window, door, street sign (ambiguous and difficult to label), plate, desk (due to confusion with bowl and dining table, respectively) and blender, hair brush (too few instances). We may add segmentations for some of these categories in the cumulative 2015 release.

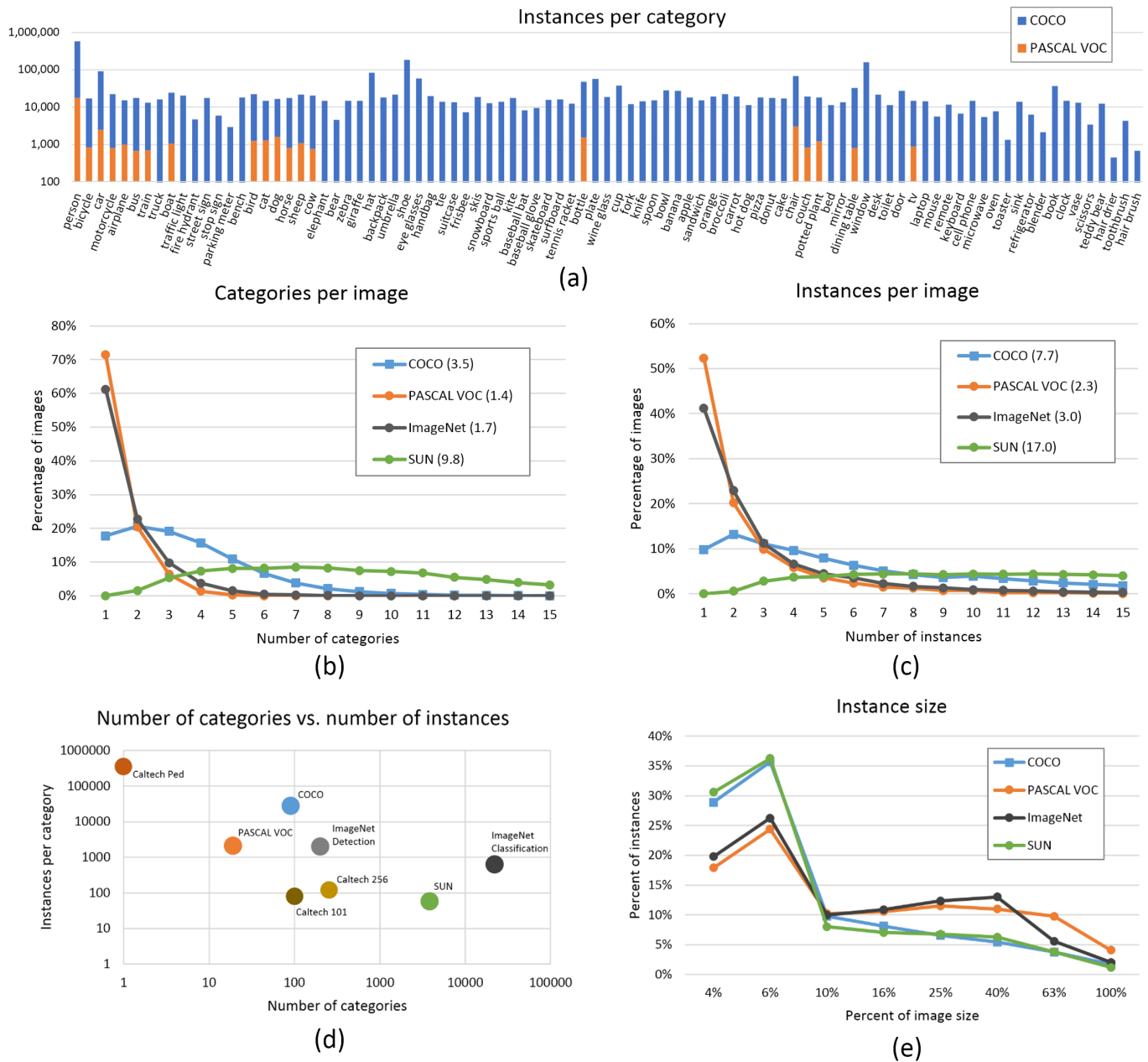


图5: (a) MS COCO和PASCAL VOC各类别的标注实例数量。(b,c) 分别为MS COCO、ImageNet Detection、PASCAL VOC和SUN数据集中每张图像的标注类别数量和标注实例数量(括号内显示平均类别数和实例数)。(d) 多个常用物体识别数据集的类别数量与每类别实例数量的关系。(e) MS COCO、ImageNet Detection、PASCAL VOC和SUN数据集的实例尺寸分布。

我们特意通过显式移除近似重复图像(使用[43]检测)并按摄影师和拍摄日期分组,以最小化不同分割间存在近似重复图像的可能性。

根据既定协议,训练和验证数据的标注将会公开,但测试数据不会。我们目前正在最终确定用于测试集自动评估的评测服务器。一旦评估完成,将补充完整的评估指标讨论。

服务器已就绪。

请注意,我们将2014年的发布版本限制在80个类别的一个子集内。对于以下11个类别,我们没有收集分割标注:帽子、鞋子、眼镜(实例过多)、镜子、窗户、门、路牌(标注模糊且困难)、盘子、书桌(分别因与碗和餐桌易混淆)、以及搅拌机、发刷(实例过少)。我们可能会在2015年的累积发布版本中为其中部分类别添加分割标注。

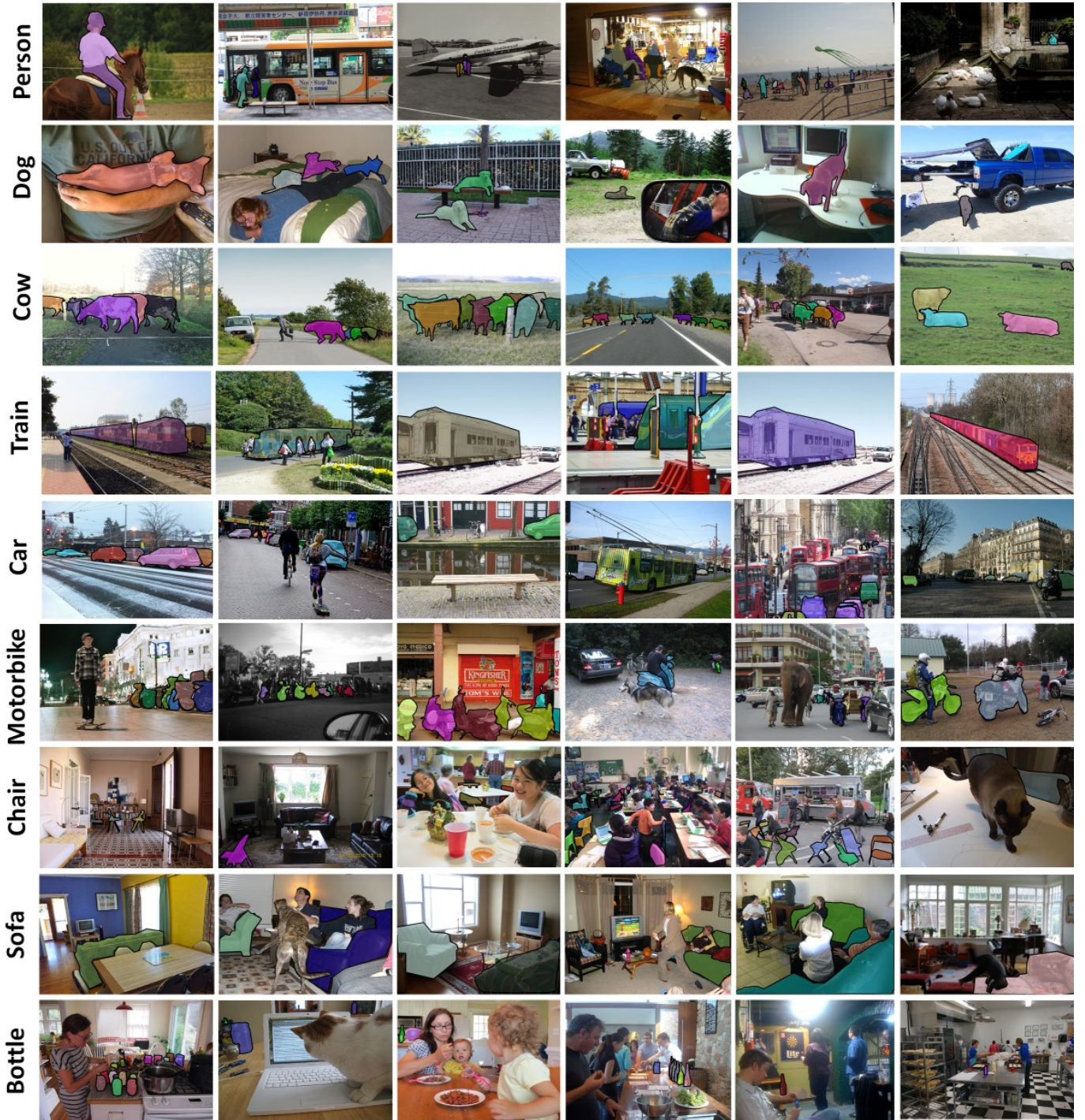


Fig. 6: Samples of annotated images in the MS COCO dataset.

7 ALGORITHMIC ANALYSIS

Bounding-box detection For the following experiments we take a subset of 55,000 images from our dataset¹ and obtain tight-fitting bounding boxes from the annotated segmentation masks. We evaluate models tested on both MS COCO and PASCAL, see Table 1. We evaluate two different models. **DPMv5-P**: the latest implementation

1. These preliminary experiments were performed before our final split of the dataset into train, val, and test. Baselines on the actual test set will be added once the evaluation server is complete.

of [44] (release 5 [45]) trained on PASCAL VOC 2012. **DPMv5-C**: the same implementation trained on COCO (5000 positive and 10000 negative images). We use the default parameter settings for training COCO models.

If we compare the average performance of DPMv5-P on PASCAL VOC and MS COCO, we find that average performance on MS COCO drops by nearly a *factor of 2*, suggesting that MS COCO does include more difficult (non-iconic) images of objects that are partially occluded, amid clutter, etc. We notice a similar drop in performance

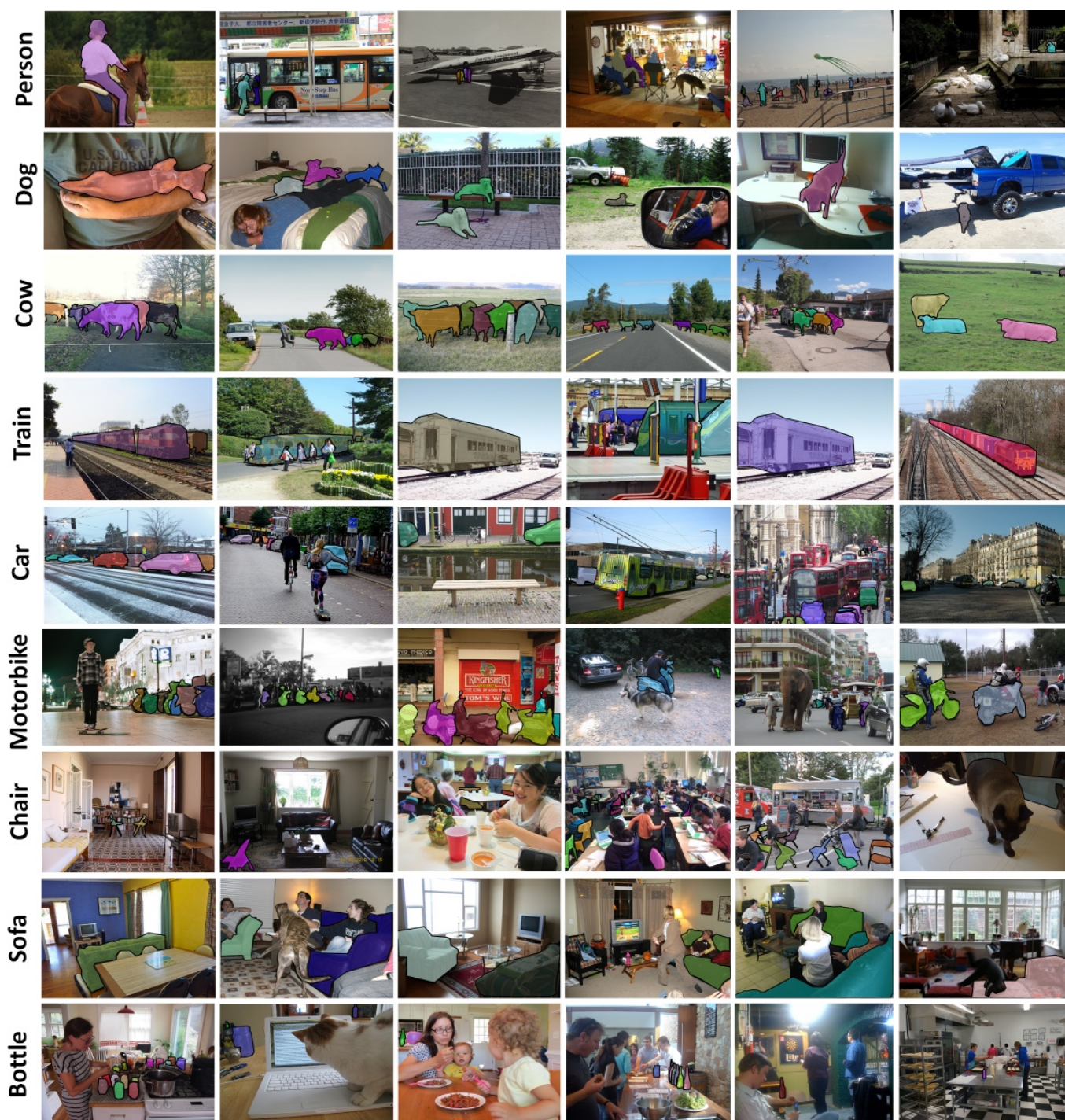


图6: MS COCO数据集中带标注的图像样本

数据集。

7 算法分析

边界框检测 在接下来的实验中, 我们从数据集¹中选取了5 5,000张图像的子集, 并通过标注的分割掩码获取紧密贴合边界框。我们评估了在MS COCO和PASCAL上测试的模型, 详见表1。我们评估了两种不同的模型: DPMv5-P (最新实现版本)

1. These preliminary experiments were performed before our final split of the dataset into train, val, and test. Baselines on the actual test set will be added once the evaluation server is complete.

来自[44] (第5版[45]) 在PASCAL VOC 2012上训练的模型。DPMv5-C: 在COCO数据集 (5000张正样本和10000张负样本图像) 上训练的相同实现。我们使用默认参数设置来训练COCO模型。

如果我们比较DPMv5-P在PASCAL VOC和MS COCO上的平均性能, 会发现其在MS COCO上的平均性能下降了近factor of 2, 这表明MS COCO确实包含了更多困难 (非标志性) 的目标图像, 这些目标存在部分遮挡、处于杂乱背景等情况。我们注意到性能出现了类似的下降。

	plane	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	moto	person	plant	sheep	sofa	train	tv	avg.
DPMv5-P	45.6	49.0	11.0	11.6	27.2	50.5	43.1	23.6	17.2	23.2	10.7	20.5	42.5	44.5	41.3	8.7	29.0	18.7	40.0	34.5	29.6
DPMv5-C	43.7	50.1	11.8	2.4	21.4	60.1	35.6	16.0	11.4	24.8	5.3	9.4	44.5	41.0	35.8	6.3	28.3	13.3	38.8	36.2	26.8
DPMv5-P	35.1	17.9	3.7	2.3	7	45.4	18.3	8.6	6.3	17	4.8	5.8	35.3	25.4	17.5	4.1	14.5	9.6	31.7	27.9	16.9
DPMv5-C	36.9	20.2	5.7	3.5	6.6	50.3	16.1	12.8	4.5	19.0	9.6	4.0	38.2	29.9	15.9	6.7	13.8	10.4	39.2	37.9	19.1

TABLE 1: **Top:** Detection performance evaluated on **PASCAL VOC 2012**. DPMv5-P is the performance reported by Girshick et al. in VOC release 5. DPMv5-C uses the same implementation, but is trained with MS COCO. **Bottom:** Performance evaluated on **MS COCO** for DPM models trained with PASCAL VOC 2012 (DPMv5-P) and MS COCO (DPMv5-C). For DPMv5-C we used 5000 positive and 10000 negative training examples. While MS COCO is considerably more challenging than PASCAL, use of more training data coupled with more sophisticated approaches [5], [6], [7] should improve performance substantially.

for the model trained on MS COCO (DPMv5-C).

The effect on detection performance of training on PASCAL VOC or MS COCO may be analyzed by comparing DPMv5-P and DPMv5-C. They use the same implementation with different sources of training data. Table 1 shows DPMv5-C still outperforms DPMv5-P in 6 out of 20 categories when testing on PASCAL VOC. In some categories (e.g., dog, cat, people), models trained on MS COCO perform worse, while on others (e.g., bus, tv, horse), models trained on our data are better.

Consistent with past observations [46], we find that including difficult (non-iconic) images during training may not always help. Such examples may act as noise and pollute the learned model if the model is not rich enough to capture such appearance variability. Our dataset allows for the exploration of such issues.

Torralba and Efros [42] proposed a metric to measure cross-dataset generalization which computes the ‘performance drop’ for models that train on one dataset and test on another. The performance difference of the DPMv5-P models across the two datasets is 12.7 AP while the DPMv5-C models only have 7.7 AP difference. Moreover, overall performance is much lower on MS COCO. These observations support two hypotheses: 1) MS COCO is significantly more difficult than PASCAL VOC and 2) models trained on MS COCO can generalize better to easier datasets such as PASCAL VOC given more training data. To gain insight into the differences between the datasets, see the appendix for visualizations of person and chair examples from the two datasets.

Generating segmentations from detections We now describe a simple method for generating object bounding boxes and segmentation masks, following prior work that produces segmentations from object detections [47], [48], [49], [50]. We learn aspect-specific pixel-level segmentation masks for different categories. These are readily learned by averaging together segmentation masks from aligned training instances. We learn different masks corresponding to the different mixtures in our DPM detector. Sample masks are visualized in Fig. 7.

Detection evaluated by segmentation Segmentation is a challenging task even assuming a detector reports correct results as it requires fine localization of object part

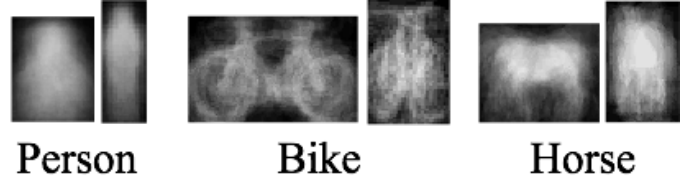


Fig. 7: We visualize our mixture-specific shape masks. We paste thresholded shape masks on each candidate detection to generate candidate segments.

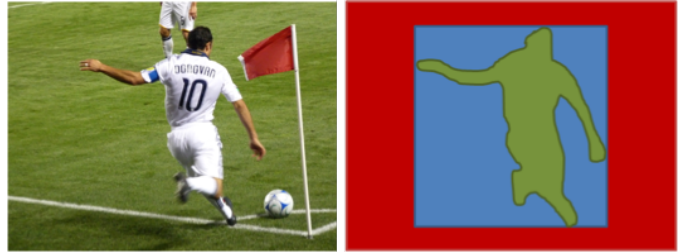


Fig. 8: Evaluating instance detections with segmentation masks versus bounding boxes. Bounding boxes are a particularly crude approximation for articulated objects; in this case, the majority of the pixels in the (blue) tight-fitting bounding-box do not lie on the object. Our (green) instance-level segmentation masks allows for a more accurate measure of object detection and localization.

boundaries. To decouple segmentation evaluation from detection correctness, we benchmark segmentation quality using only correct detections. Specifically, given that the detector reports a correct bounding box, how well does the predicted segmentation of that object match the ground truth segmentation? As criterion for correct detection, we impose the standard requirement that intersection over union between predicted and ground truth boxes is at least 0.5. We then measure the intersection over union of the predicted and ground truth segmentation masks, see Fig. 8. To establish a baseline for our dataset, we project learned DPM part masks onto the image to create segmentation masks. Fig. 9 shows results of this segmentation baseline for the DPM learned on the 20 PASCAL categories and tested on our dataset.

	plane	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	moto	person	plant	sheep	sofa	train	tv	avg.
DPMv5-P	45.6	49.0	11.0	11.6	27.2	50.5	43.1	23.6	17.2	23.2	10.7	20.5	42.5	44.5	41.3	8.7	29.0	18.7	40.0	34.5	29.6
DPMv5-C	43.7	50.1	11.8	2.4	21.4	60.1	35.6	16.0	11.4	24.8	5.3	9.4	44.5	41.0	35.8	6.3	28.3	13.3	38.8	36.2	26.8
DPMv5-P	35.1	17.9	3.7	2.3	7	45.4	18.3	8.6	6.3	17	4.8	5.8	35.3	25.4	17.5	4.1	14.5	9.6	31.7	27.9	16.9
DPMv5-C	36.9	20.2	5.7	3.5	6.6	50.3	16.1	12.8	4.5	19.0	9.6	4.0	38.2	29.9	15.9	6.7	13.8	10.4	39.2	37.9	19.1

表1: 上: 在PASCAL VOC 2012上评估的检测性能。DPMv5-P是Girshick等人在VOC第5版发布中报告的性能。DPMv5-C采用相同实现, 但使用MS COCO进行训练。下: 在MS COCO上评估使用PASCAL VOC 2012 (DPMv5-P) 和MS COCO (DPMv5-C) 训练的DPM模型的性能。对于DPMv5-C, 我们使用了5000个正例和10000个负例训练样本。尽管MS COCO比PASCAL更具挑战性, 但结合更多训练数据及更先进的方法[5]、[6]、[7]应能显著提升性能。

对于在MS COCO上训练的模型 (DPMv5-C)。

通过比较DPMv5-P和DPMv5-C, 可以分析在PASCAL VOC或MS COCO上训练对检测性能的影响。它们采用相同的实现方式, 但训练数据来源不同。表1显示, 在PASCAL VOC上进行测试时, DPMv5-C仍在20个类别中的6个类别上优于DPMv5-P。在某些类别 (如狗、猫、人) 上, 基于MS COCO训练的模型表现较差; 而在其他类别 (如公交车、电视、马) 上, 基于我们数据训练的模型表现更佳。

与过去的观察一致[46], 我们发现训练中包含困难 (非标志性) 图像可能并不总是有帮助。如果模型不够丰富, 无法捕捉这种外观变化, 这类示例可能起到噪声作用, 污染学习到的模型。我们的数据集允许对此类问题进行探索。

Torrallba和Efros[42]提出了一种衡量跨数据集泛化能力的指标, 该指标通过计算模型在一个数据集上训练、在另一个数据集上测试时的“性能下降”来评估。DPMv5-P模型在两个数据集间的性能差异为12.7 AP, 而DPMv5-C模型的差异仅为7.7 AP。此外, 模型在MS COCO上的整体性能明显更低。这些观察结果支持了两个假设: 1) MS COCO的难度显著高于PASCAL VOC; 2) 在更多训练数据支持下, 基于MS COCO训练的模型能更好地泛化到如PASCAL VOC等较简单的数据集。为深入理解数据集间的差异, 可参阅附录中两个数据集关于“人”和“椅子”类别的可视化对比示例。

从检测结果生成分割 我们现在描述一种生成物体边界框和分割掩码的简单方法, 该方法遵循先前从物体检测生成分割的研究[47]、[48]、[49]、[50]。我们为不同类别学习特定长宽比的像素级分割掩码。这些掩码可以通过对齐训练实例的分割掩码进行平均来轻松学习。我们学习了与DPM检测器中不同混合成分对应的不同掩码。示例掩码在图7中可视化。

基于分割的检测评估 即使假设检测器报告了正确结果, 分割仍是一项具有挑战性的任务, 因为它需要对物体部件进行精细定位。

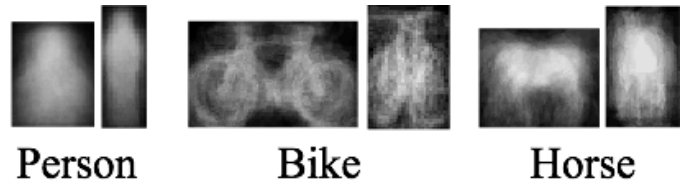


图7: 我们可视化我们的混合特定形状掩码。我们将阈值化后的形状掩码粘贴在每个候选检测框上以生成候选分割区域。

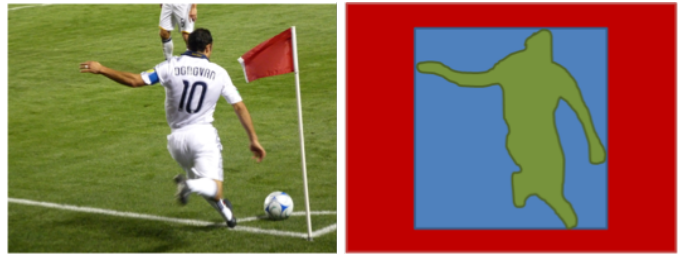


图8: 使用分割掩码与边界框评估实例检测。对于铰接式物体, 边界框是一种特别粗略的近似; 在此案例中, (蓝色) 紧密贴合边界框内的大部分像素并不位于物体上。我们的(绿色)实例级分割掩码能够为物体检测与定位提供更精确的度量。

边界。为了将分割评估与检测正确性解耦, 我们仅使用正确的检测结果来基准测试分割质量。具体而言, 给定检测器报告了一个正确的边界框, 该物体的预测分割与真实分割的匹配程度如何? 作为正确检测的标准, 我们施加了预测框与真实框的交并比至少为0.5的标准要求。随后, 我们测量预测分割掩码与真实分割掩码的交并比, 见图8。为了为我们的数据集建立基线, 我们将学习到的DPM部件掩码投影到图像上以创建分割掩码。图9展示了在20个PASCAL类别上学习并测试于我们数据集的DPM分割基线结果。

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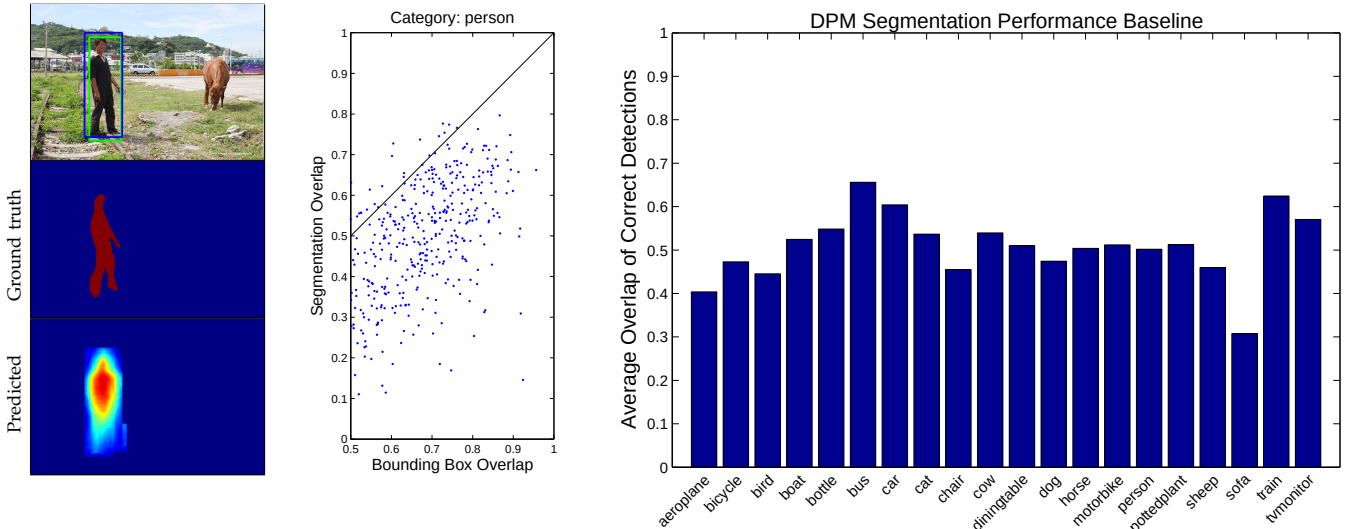


Fig. 9: A predicted segmentation might not recover object detail even though detection and ground truth bounding boxes overlap well (left). Sampling from the person category illustrates that predicting segmentations from top-down projection of DPM part masks is difficult even for correct detections (center). Average segmentation overlap measured on MS COCO for the 20 PASCAL VOC categories demonstrates the difficulty of the problem (right).

8 DISCUSSION

We introduced a new dataset for detecting and segmenting objects found in everyday life in their natural environments. Utilizing over 70,000 worker hours, a vast collection of object instances was gathered, annotated and organized to drive the advancement of object detection and segmentation algorithms. Emphasis was placed on finding non-iconic images of objects in natural environments and varied viewpoints. Dataset statistics indicate the images contain rich contextual information with many objects present per image.

There are several promising directions for future annotations on our dataset. We currently only label “things”, but labeling “stuff” may also provide significant contextual information that may be useful for detection. Many object detection algorithms benefit from additional annotations, such as the amount an instance is occluded [4] or the location of keypoints on the object [10]. Finally, our dataset could provide a good benchmark for other types of labels, including scene types [3], attributes [9], [8] and full sentence written descriptions [51]. We are actively exploring adding various such annotations.

To download and learn more about MS COCO please see the project website². MS COCO will evolve and grow over time; up to date information is available online.

Acknowledgments Funding for all crowd worker tasks was provided by Microsoft. P.P. and D.R. were supported by ONR MURI Grant N00014-10-1-0933. We would like to thank all members of the community who provided valuable feedback throughout the process of defining and collecting the dataset.

APPENDIX OVERVIEW

In the appendix, we provide detailed descriptions of the AMT user interfaces and the full list of 272 candidate categories (from which our final 91 were selected) and 40 scene categories (used for scene-object queries).

APPENDIX I: USER INTERFACES

We describe and visualize our user interfaces for collecting non-iconic images, category labeling, instance spotting, instance segmentation, segmentation verification and finally crowd labeling.

Non-iconic Image Collection Flickr provides a rich image collection associated with text captions. However, captions might be inaccurate and images may be iconic. To construct a high-quality set of non-iconic images, we first collected candidate images by searching for pairs of object categories, or pairs of object and scene categories. We then created an AMT filtering task that allowed users to remove invalid or iconic images from a grid of 128 candidates, Fig. 10. We found the choice of instructions to be crucial, and so provided users with examples of iconic and non-iconic images. Some categories rarely co-occurred with others. In such cases, we collected candidates using only the object category as the search term, but apply a similar filtering step, Fig. 10(b).

Category Labeling Fig. 12(a) shows our interface for category labeling. We designed the labeling task to encourage workers to annotate all categories present in the image. Workers annotate categories by dragging and dropping icons from the bottom category panel onto a corresponding object instance. Only a single instance of each object category needs to be annotated in the image. We group icons by the super-categories from Fig. 11, allowing workers to quickly skip categories that are unlikely to be present.

2. <http://mscoco.org/>

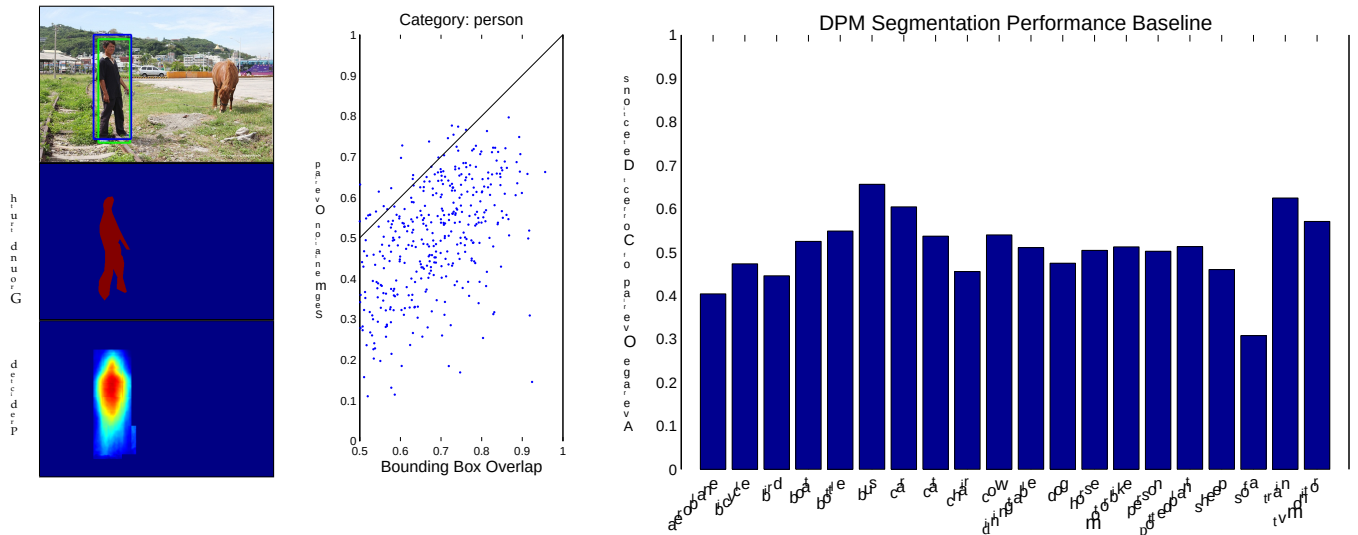


图9: 即使检测框与真实标注框重叠良好, 预测的分割结果也可能无法恢复物体细节(左)。从人物类别中采样显示, 即使对于正确的检测, 从DPM部件掩码自上而下投影预测分割也十分困难(中)。在MS COCO数据集上对20个PASCAL VOC类别测量的平均分割重叠率, 证明了该问题的难度(右)。

8 讨论

我们引入了一个新的数据集, 用于检测和分割自然环境中日常物体的发现。通过超过70,000个工时的投入, 我们收集、标注并组织了大量物体实例, 以推动物体检测和分割算法的发展。重点在于寻找自然环境中物体的非标志性图像及多样化的视角。数据集统计显示, 图像包含丰富的上下文信息, 每张图像中呈现多个物体。

对于未来在我们数据集上的标注工作, 有几个前景广阔的方向。目前我们仅标注了“物体”, 但标注“背景材质”同样可能提供重要的上下文信息, 这对检测任务或有助益。许多目标检测算法都能从额外标注中受益, 例如实例被遮挡的程度[4]或物体关键点的位置[10]。最后, 我们的数据集也可为其他类型的标注提供优质基准, 包括场景类型[3]、属性[9][8]以及完整的句子描述[51]。我们正在积极探索添加此类多样化标注的可能性。

要下载并了解更多关于MS COCO的信息, 请访问项目网站²。MS COCO将随时间不断演进和发展; 最新信息可在网上获取。

致谢 所有众包任务的资金由微软提供P.P.和D.R.得到了ONR MURI资助(编号N00014-10-1-0933)的支持。我们衷心感谢在数据集定义和收集过程中提供宝贵反馈的社区全体成员。

附录概述

在附录中, 我们详细描述了AMT用户界面, 并提供了272个候选类别(我们最终从中选取了91个)和40个场景类别(用于场景-对象查询)的完整列表。

附录一: 用户界面

我们描述并展示了用于收集非标志性图像、类别标注、实例定位、实例分割、分割验证以及最终众包标注的用户界面。

非标志性图像收集 Flickr 提供了丰富的带有文字说明的图像集合。然而, 说明可能不准确, 图像也可能是标志性的。为了构建高质量的非标志性图像集, 我们首先通过搜索成对的物体类别, 或物体与场景类别的组合来收集候选图像。随后, 我们设计了一个 AMT 筛选任务, 允许用户从 128 个候选图像的网格中移除无效或标志性图像, 如图 10 所示。我们发现指令的选择至关重要, 因此为用户提供了标志性和非标志性图像的示例。某些类别很少与其他类别同时出现。在这种情况下, 我们仅使用物体类别作为搜索词来收集候选图像, 但应用了类似的筛选步骤, 如图 10 (b) 所示。

类别标注 图(a)展示了我们的类别标注界面。我们将标注任务设计为鼓励标注者为图像中出现的所有类别添加标注。标注者通过将底部类别面板中的图标拖放到对应的物体实例上来完成类别标注。每个物体类别在图像中仅需标注一个实例。我们按照图11中的超类别对图标进行分组, 使标注者能快速跳过不太可能出现的类别。

2. <http://mscoco.org/>



Fig. 10: User interfaces for non-iconic image collection. (a) Interface for selecting non-iconic images containing pairs of objects. (b) Interface for selecting non-iconic images for categories that rarely co-occurred with others.

Instance Spotting Fig. 12(b) depicts our interface for labeling all instances of a given category. The interface is initialized with a blinking icon specifying a single instance obtained from the previous category-labeling stage. Workers are then asked to spot and click on up to 10 total instances of the given category, placing a single cross anywhere within the region of each instance. In order to spot small objects, we found it crucial to include a “magnifying glass” feature that doubles the resolution of a worker’s currently selected region.

Instance Segmentation Fig. 12(c) shows our user interface for instance segmentation. We modified source code from the OpenSurfaces project [16], which defines a single AMT task for segmenting multiple regions of a homogenous material in real-scenes. In our case, we define a single task for segmenting a single object instance labeled from the previous annotation stage. To aid the segmentation process, we added a visualization of the object category icon to remind workers of the category to be segmented. Crucially, we also added zoom-in functionality to allow for efficient annotation of small objects and curved boundaries. In the previous annotation stage, to ensure high coverage of all object instances, we used multiple workers to label all instances per image. We would like to segment *all* such object instances, but instance annotations across different workers may refer to different or redundant instances. To resolve this correspondence ambiguity, we sequentially post AMT segmentation tasks, ignoring instance annotations that are already covered by an existing segmentation mask.

Segmentation Verification Fig. 12(d) shows our user interface for segmentation verification. Due to the time consuming nature of the previous task, each object instance is segmented only once. The purpose of the veri-

fication stage is therefore to ensure that each segmented instance from the previous stage is of sufficiently high quality. Workers are shown a grid of 64 segmentations and asked to select poor quality segmentations. Four of the 64 segmentation are known to be bad; a worker must identify 3 of the 4 known bad segmentations to complete the task. Each segmentation is initially shown to 3 annotators. If any of the annotators indicates the segmentation is bad, it is shown to 2 additional workers. At this point, any segmentation that doesn’t receive at least 4 of 5 favorable votes is discarded and the corresponding instance added back to the pool of unsegmented objects. Examples of borderline cases that either passed (4/5 votes) or were rejected (3/5 votes) are shown in Fig. 15.

Crowd Labeling Fig. 12(e) shows our user interface for crowd labeling. As discussed, for images containing ten object instances or fewer of a given category, every object instance was individually segmented. In some images, however, the number of instances of a given category is much higher. In such cases crowd labeling provided a more efficient method for annotation. Rather than requiring workers to draw exact polygonal masks around each object instance, we allow workers to “paint” all pixels belonging to the category in question. Crowd labeling is similar to semantic segmentation as object instance are not individually identified. We emphasize that crowd labeling is only necessary for images containing more than ten object instances of a given category.

APPENDIX II: OBJECT & SCENE CATEGORIES

Our dataset contains 91 object categories (the 2014 release contains segmentation masks for 80 of these categories). We began with a list of frequent object categories taken from WordNet, LabelMe, SUN and other sources as well as categories derived from a free recall experiment with young children. The authors then voted on the resulting 272 categories with the aim of sampling a diverse and computationally challenging set of categories; see §3 for details. The list in Table 2 enumerates those 272 categories in descending order of votes. As discussed, the final selection of 91 categories attempts to pick categories with high votes, while keeping the number of categories per super-category (animals, vehicles, furniture, etc.) balanced.

As discussed in §3, in addition to using object-object queries to gather non-iconic images, object-scene queries also proved effective. For this task we selected a subset of 40 scene categories from the SUN dataset that frequently co-occurred with object categories of interest. Table 3 enumerates the 40 scene categories (evenly split between indoor and outdoor scenes).

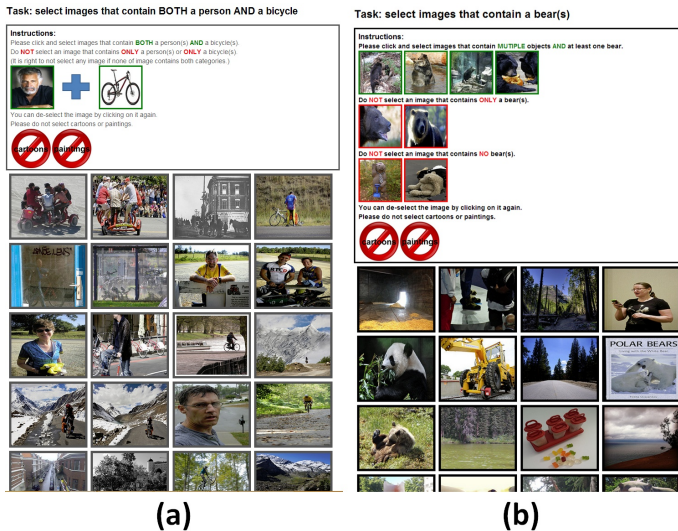


图10：非标志性图像收集的用户界面。(a) 用于选择包含成对对象的非标志性图像的界面。(b) 用于选择与其他类别极少共现的类别的非标志性图像的界面。

实例标注 图(b)展示了我们用于标注给定类别所有实例的界面。该界面初始化时会显示一个闪烁图标，该图标代表从上一阶段类别标注中获取的单个实例。随后，工作人员需识别并点击该类别最多10个实例，在每个实例区域内任意位置标记一个十字。为准确识别小型物体，我们加入了“放大镜”功能，可将工作人员当前选定区域的分辨率提升一倍，这一功能至关重要。

实例分割 图12(c)展示了我们用于实例分割的用户界面。我们修改了OpenSurfaces项目[16]的源代码，该项目为真实场景中同质材料的多个区域分割定义了一个单一的AMT任务。在我们的案例中，我们定义了一个单一任务，用于分割先前标注阶段标记出的单个对象实例。为了辅助分割过程，我们添加了对对象类别图标可视化，以提醒工作人员需要分割的类别。关键的是，我们还添加了放大功能，以便高效标注小物体和弯曲边界。在之前的标注阶段，为确保所有对象实例的高覆盖率，我们使用多名工作人员对每张图像中的所有实例进行标注。我们希望分割all个这样的对象实例，但不同工作人员的实例标注可能指向不同或冗余的实例。为解决这种对应歧义，我们按顺序发布AMT分割任务，忽略已被现有分割掩码覆盖的实例标注。

分割验证图12(d)展示了我们用于分割验证的用户界面。由于先前任务耗时较长，每个对象实例仅被分割一次。验证的目的在于

因此，质量验证阶段旨在确保前一阶段分割出的每个实例都具有足够高的质量。工作人员会看到一个包含64个分割结果的网格，并被要求选择质量较差的分割。其中已知有4个分割质量不佳；工作人员必须识别出这4个已知不良分割中的3个才能完成任务。每个分割最初会展示给3位标注者。如果任何标注者认为分割质量差，该分割会再展示给另外2位工作人员。此时，任何未获得至少4票（共5票）支持的分割将被丢弃，对应的实例将重新放回未分割对象池中。图15展示了部分边缘案例，它们要么获得通过（4/5票），要么被拒绝（3/5票）。

众包标注 图(e)展示了我们用于众包标注的用户界面。如前所述，对于包含十个或更少特定类别物体实例的图像，每个物体实例均被单独分割标注。然而在某些图像中，特定类别的实例数量远超于此。此类情况下，众包标注提供了更高效的标注方式：我们不再要求标注者为每个物体实例绘制精确的多边形掩膜，而是允许标注者以“涂绘”方式标注该类别所有像素。众包标注类似于语义分割，因为物体实例并未被单独区分。需要强调的是，仅当图像中特定类别的物体实例超过十个时，才需采用众包标注方式。

附录二：物体与场景类别

我们的数据集包含91个对象类别（2014年版本为其中80个类别提供了分割掩码）。我们最初从WordNet、LabelMe、SUN等来源以及针对幼童的自由回忆实验衍生出的类别中，选取了常见对象类别清单。随后作者们对由此产生的272个类别进行投票，旨在筛选出多样化且具有计算挑战性的类别集合；详见§3。表2按得票降序列举了这272个类别。如前所述，最终选定的91个类别在尽量保持每个超类别（动物、交通工具、家具等）数量平衡的前提下，优先选择了得票较高的类别。

如§3所述，除了使用物体-物体查询来收集非标志性图像外，物体-场景查询也被证明是有效的。为此任务，我们从SUN数据集中选取了40个场景类别的子集，这些场景类别常与我们感兴趣的对象类别共同出现。表3列举了这40个场景类别（室内和室外场景各占一半）。

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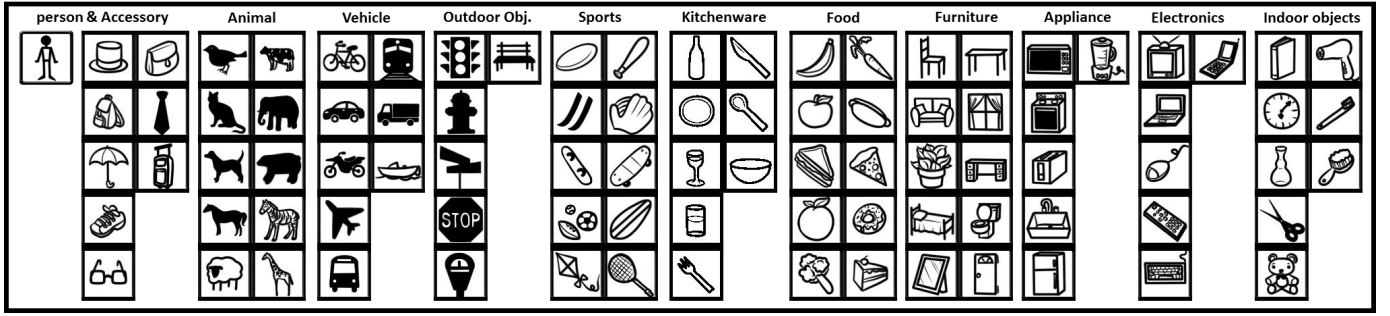


Fig. 11: Icons of 91 categories in the MS COCO dataset grouped by 11 super-categories. We use these icons in our annotation pipeline to help workers quickly reference the indicated object category.

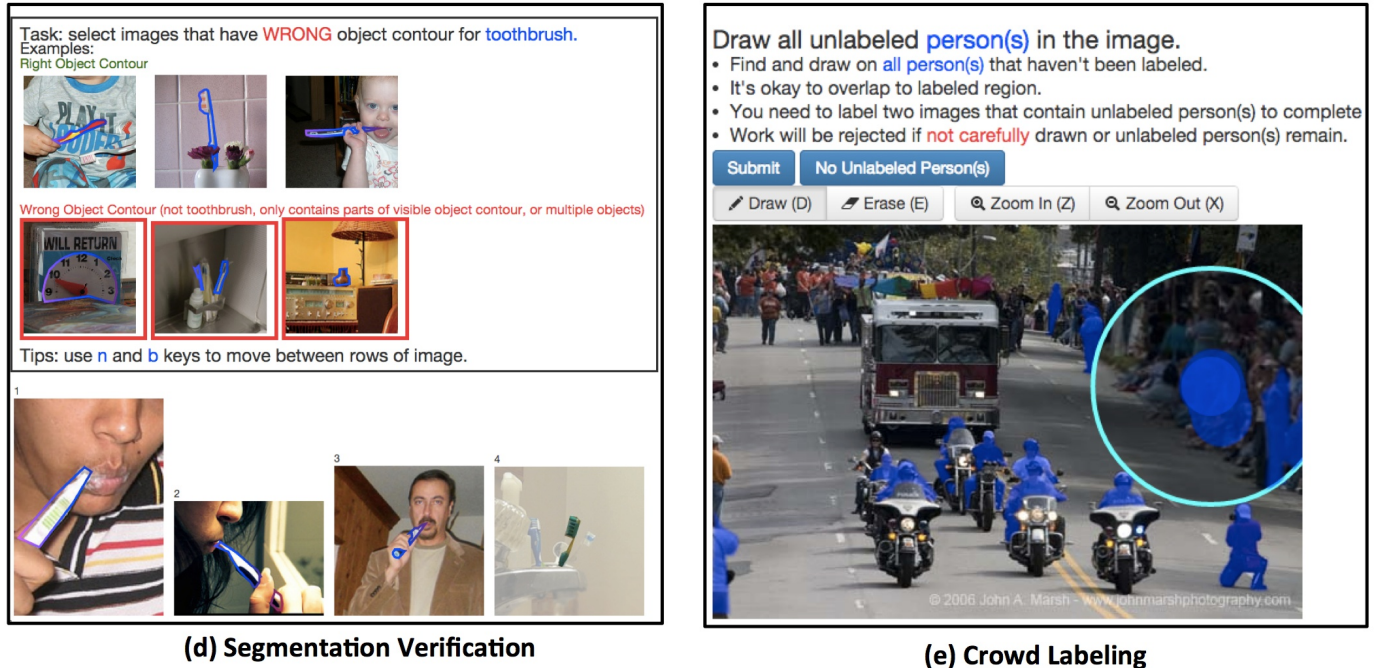


Fig. 12: User interfaces for collecting instance annotations, see text for details.

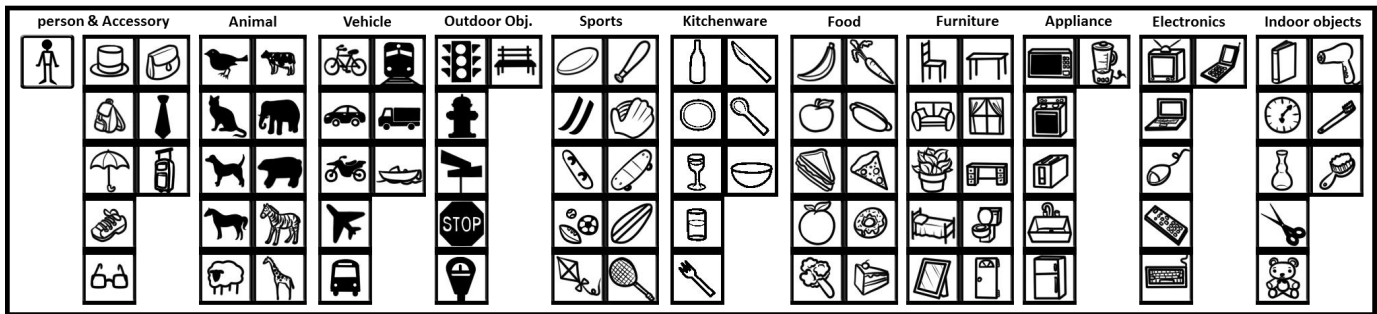


图. 11: MS COCO 数据集中 91 个类别的图标按 11 个超类别分组。我们在研究中使用了这些图标 anno 标注流水线，帮助工人快速参考指定的对象类别。

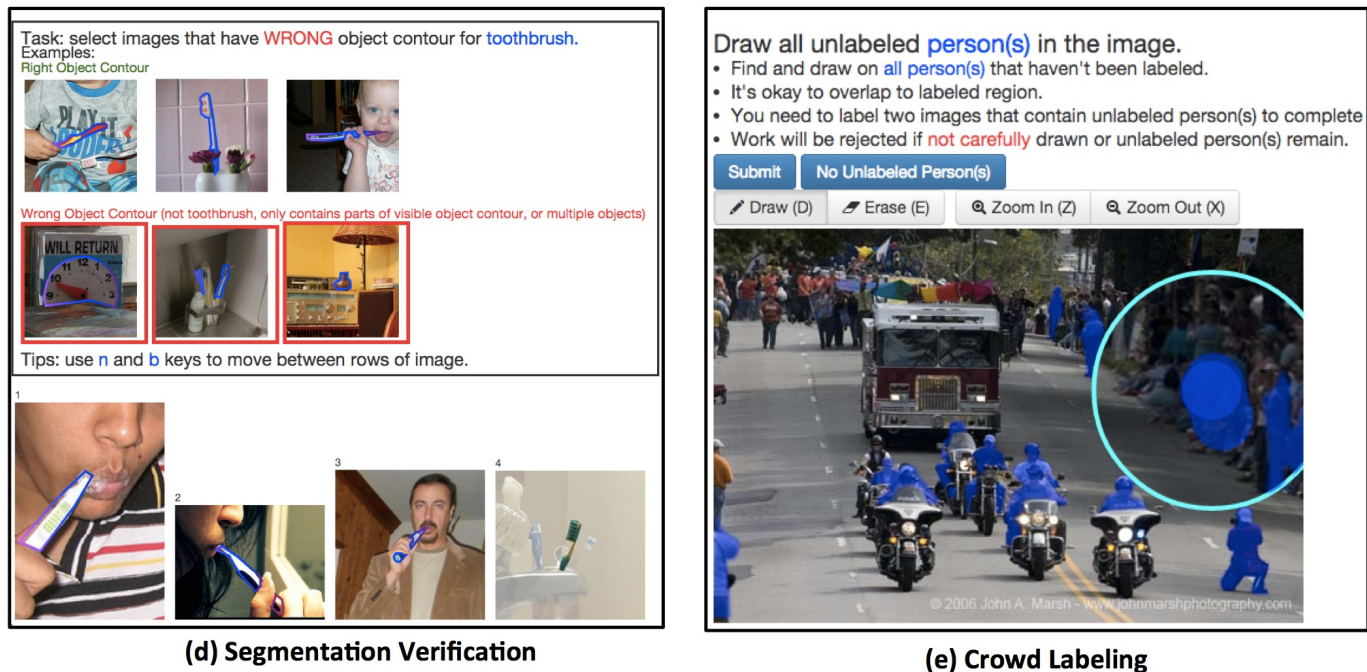
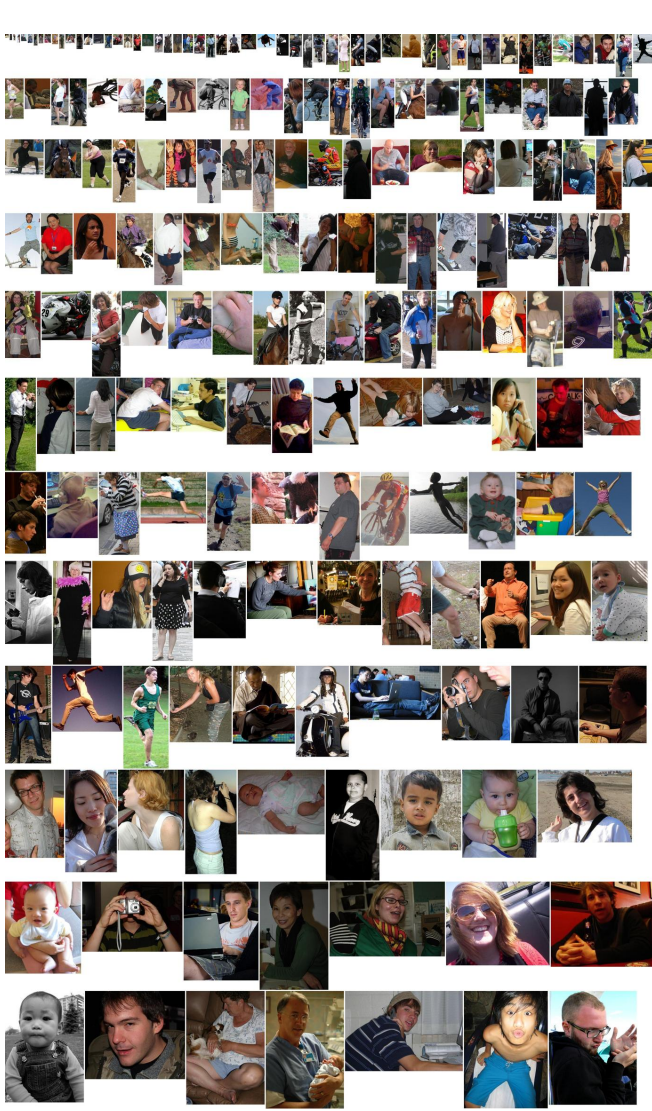


图12: 用于收集实例标注的用户界面，详见正文说明。



(a) PASCAL VOC.

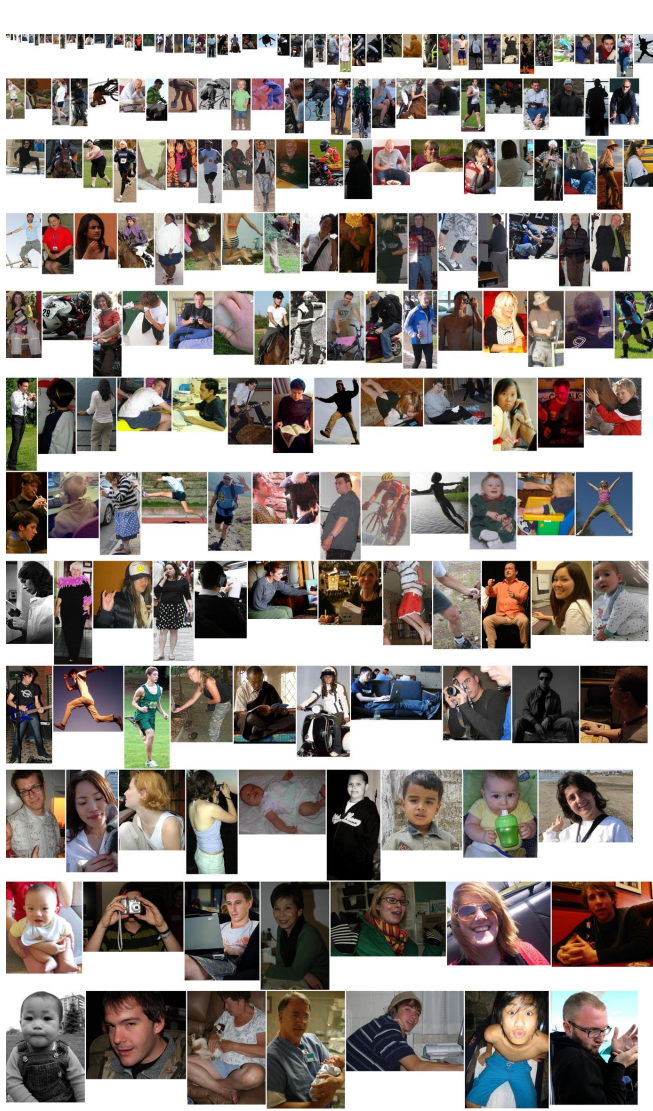


(b) MS COCO.

Fig. 13: Random person instances from PASCAL VOC and MS COCO. At most one instance is sampled per image.

person	bicycle	car	motorcycle	bird	cat	dog	horse	sheep	bottle
chair	couch	potted plant	tv	cow	airplane	hat*	license plate	bed	laptop
fridge	microwave	sink	oven	toaster	bus	train	mirror*	dining table	elephant
banana	bread	toilet	book	boat	plate*	cell phone	mouse	remote	clock
face	hand	apple	keyboard	backpack	steering wheel	wine glass	chicken	zebra	shoe*
eye	mouth	scissors	truck	traffic light	eyeglasses*	cup	blender*	hair drier	wheel
street sign*	umbrella	door*	fire hydrant	bowl	teapot	fork	knife	spoon	bear
headlights	window*	desk*	computer	refrigerator	pizza	squirrel	duck	frisbee	guitar
nose	teddy bear	tie	stop sign	surfboard	sandwich	pen/pencil	kite	orange	toothbrush
printer	pans	head	sports ball	broccoli	suitcase	carrot	chandelier	parking meter	fish
handbag	hot dog	stapler	basketball hoop	donut	vase	baseball bat	baseball glove	giraffe	jacket
skis	snowboard	table lamp	egg	door handle	power outlet	hair	tiger	table	coffee table
skateboard	helicopter	tomato	tree	bunny	pillow	tennis racket	cake	feet	bench
chopping board	washer	lion	monkey	hair brush*	light switch	arms	legs	house	cheese
goat	magazine	key	picture frame	cupcake	fan (ceiling/floor)	frogs	rabbit	owl	scarf
ears	home phone	pig	strawberries	pumpkin	van	kangaroo	rhinoceros	sailboat	deer
playing cards	towel	hyppo	can	dollar bill	doll	soup	meat	window	muffins
tire	necklace	tablet	corn	ladder	pineapple	candle	desktop	carpet	cookie
toy cars	bracelet	bat	balloon	gloves	milk	pants	wheelchair	building	bacon
box	platypus	pancake	cabinet	whale	dryer	torso	lizard	shirt	shorts
pasta	grapes	shark	swan	fingers	towel	side table	gate	beans	flip flops
moon	road/street	fountain	fax machine	bat	hot air balloon	cereal	seahorse	rocket	cabinets
basketball	telephone	movie (disc)	football	goose	long sleeve shirt	short sleeve shirt	raft	rooster	copier
radio	fences	goal net	toys	engine	soccer ball	field goal posts	socks	tennis net	seats
elbows	aardvark	dinosaur	unicycle	honey	legos	fly	roof	baseball	mat
ipad	iphone	hoop	hen	back	table cloth	soccer nets	turkey	pajamas	underpants
goldfish	robot	crusher	animal crackers	basketball court	horn	firefly	armpits	nectar	super hero costume
jetpack	robots								

TABLE 2: Candidate category list (272). **Bold**: selected categories (91). **Bold***: omitted categories in 2014 release (11).



(a) PASCAL VOC.

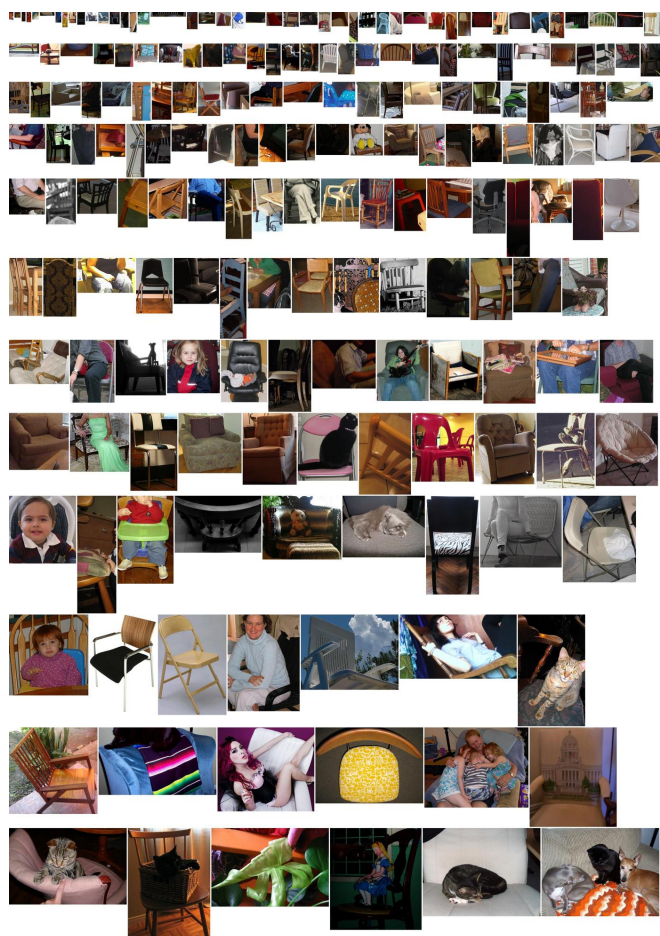


(b) MS COCO.

图13: 来自PASCAL VOC和MS COCO的随机人物实例。每张图像最多采样一个实例。

person	bicycle	car	motorcycle	bird	cat	dog	horse	sheep	bottle
chair	couch	potted plant	tv	cow	airplane	hat*	license plate	bed	laptop
fridge	microwave	sink	oven	toaster	bus	train	mirror*	dining table	elephant
banana	bread	toilet	book	boat	plate*	cell phone	mouse	remote	clock
face	hand	apple	keyboard	backpack	steering wheel	wine glass	chicken	zebra	shoe*
eye	mouth	scissors	truck	traffic light	eyeglasses*	cup	blender*	hair drier	wheel
street sign*	umbrella	door*	fire hydrant	bowl	teapot	fork	knife	spoon	bear
headlights	window*	desk*	computer	refrigerator	pizza	squirrel	duck	frisbee	guitar
nose	teddy bear	tie	stop sign	surfboard	sandwich	pen/pencil	kite	orange	toothbrush
printer	pans	head	sports ball	broccoli	suitcase	carrot	chandelier	parking meter	fish
handbag	hot dog	stapler	basketball hoop	donut	vase	baseball bat	baseball glove	giraffe	jacket
skis	snowboard	table lamp	egg	door handle	power outlet	hair	tiger	table	coffee table
skateboard	helicopter	tomato	tree	bunny	pillow	tennis racket	cake	feet	bench
chopping board	washer	lion	monkey	hair brush*	light switch	arms	legs	house	cheese
goat	magazine	key	picture frame	cupcake	fan (ceiling/floor)	frogs	rabbit	owl	scarf
ears	home phone	pig	strawberries	pumpkin	van	kangaroo	rhinoceros	sailboat	deer
playing cards	towel	hippo	can	dollar bill	doll	soup	meat	window	muffins
tire	necklace	tablet	corn	ladder	pineapple	candle	desktop	carpet	cookie
toy cars	bracelet	bat	balloon	gloves	milk	pants	wheelchair	building	bacon
box	platypus	pancake	cabinet	whale	dryer	torso	lizard	shirt	shorts
pasta	grapes	shark	swan	fingers	towel	side table	gate	beans	flip flops
moon	road/street	fountain	fax machine	bat	hot air balloon	cereal	seahorse	rocket	cabinets
basketball	telephone	movie (disc)	football	goose	long sleeve shirt	short sleeve shirt	raft	rooster	copier
radio	fences	goal net	toys	engine	soccer ball	field goal posts	socks	tennis net	seats
elbows	aardvark	dinosaur	unicycle	honey	legos	fly	roof	baseball	mat
ipad	iphone	hoop	hen	back	table cloth	soccer nets	turkey	pajamas	underpants
goldfish	robot	crusher	animal crackers	basketball court	horn	firefly	arm pits	nectar	super hero costume
jetpack	robots								

表2: 候选类别列表 (272)。粗体: 已选类别 (91)。粗体*: 2014年版本中省略的类别 (11)。



(a) PASCAL VOC.



(b) MS COCO.

Fig. 14: Random chair instances from PASCAL VOC and MS COCO. At most one instance is sampled per image.

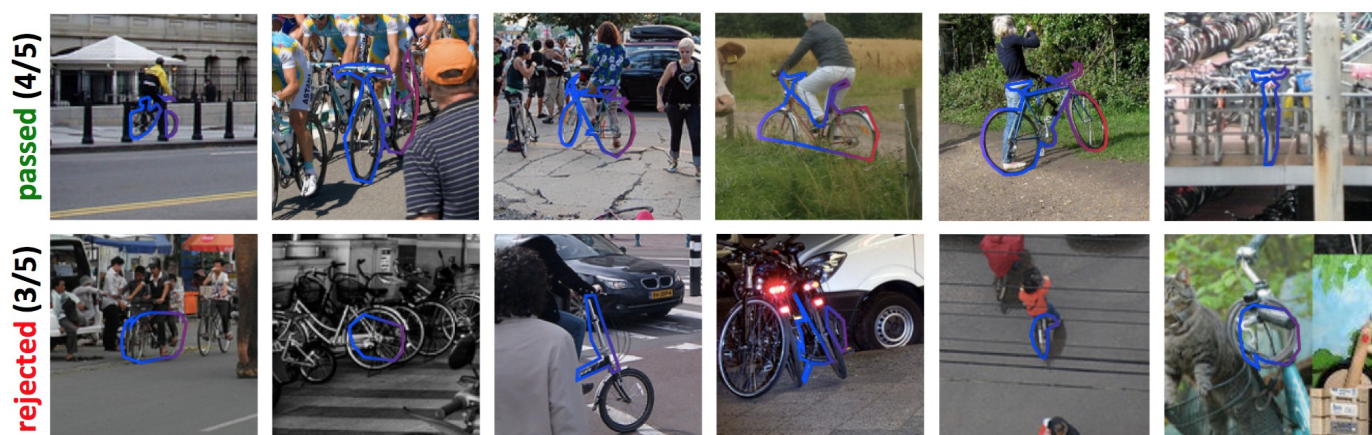
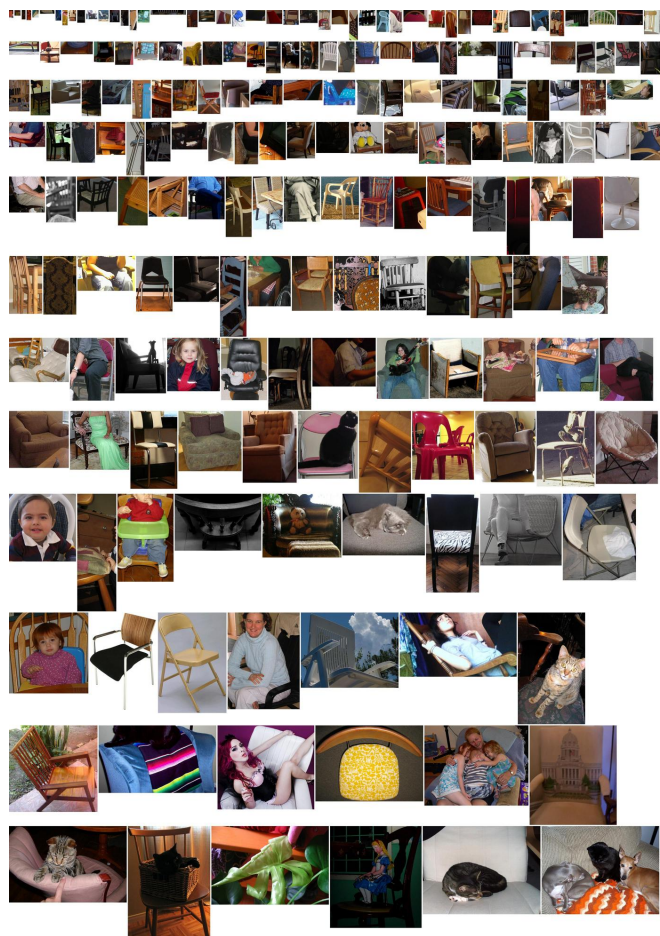


Fig. 15: Examples of borderline segmentations that passed (top) or were rejected (bottom) in the verification stage.

library	church	office	restaurant	kitchen	living room	bathroom	factory	campus	bedroom
child's room	dining room	auditorium	shop	home	hotel	classroom	cafeteria	hospital room	food court
street	park	beach	river	village	valley	market	harbor	yard	parking lot
lighthouse	railway	playground	swimming pool	forest	gas station	garden	farm	mountain	plaza

TABLE 3: Scene category list.



(a) PASCAL VOC.



(b) MS COCO.

图. 14: 来自PASCAL VOC和MS COCO的随机椅子实例。每张图像最多采样一个实例。

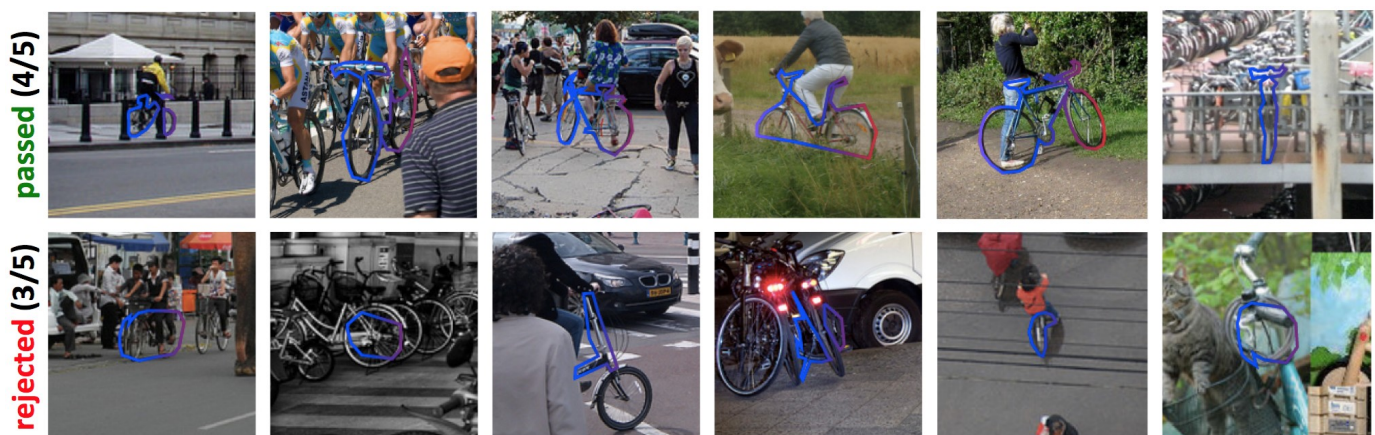


图. 15: 在验证阶段通过（顶部）或被拒绝（底部）的边界分割示例。

library	church	office	restaurant	kitchen	living room	bathroom	factory	campus	bedroom
child's room	dining room	auditorium	shop	home	hotel	classroom	cafeteria	hospital room	food court
street	park	beach	river	village	valley	market	harbor	yard	parking lot
lighthouse	railway	playground	swimming pool	forest	gas station	garden	farm	mountain	plaza

TABLE 3: Scene category list.